

Forests for Differences: complete interim simulation results for DiD-BCF

Simulation report

1 August 2026

Abstract

This report reanalyzes the complete set of currently available simulation summaries after the identification and design corrections. The main full-grid analysis covers the structured DiD-BCF specification in all available B1, B2, and D designs at linearity degrees $d = 1, 2, 3$. The B1 designs include the baseline, strong-confounder, serial-correlation, selection-on-observables, and sharp-null cases. The B2 designs contain the ordinary and serial-correlation sample-size sweeps. The D designs contain staggered adoption and its stronger contamination variant. Each available full-grid cell has 100 replications, except for the 200-replication TWFE null benchmark.

The report also analyzes the separate post-correction route file, which is the only current file containing both structured DiD-BCF and structured DiD-BCF with a unit-level random intercept. That route audit covers the B2 ordinary sweep at $d = 1$, with 15 replications per panel size and $N \in \{200, 400, 800, 1600\}$. Both specifications are called DiD-BCF throughout, with the unit-intercept column identifying the implementation.

The complete structured grid shows small scalar bias in the B1 and B2 designs, stable performance across the corrected $d = 2$ and $d = 3$ axes for the plain structured fit, and strong performance on the staggered cohort-time targets. The posterior correction improves uncertainty calibration in the serial and staggered designs at $d = 1$, but that advantage does not survive the nonlinear nuisance axis: at $d = 2$ and $d = 3$ its event-study RMSE in the staggered designs exceeds pooled TWFE at every event time. Under the sharp null both DiD-BCF specifications are correctly sized, the plain fit conservatively so. The staggered results reject the use of pooled TWFE as the main heterogeneous effect benchmark. The two covariate-adaptive benchmarks, DoubleML DR-DiD and grf-DiD (Wang), are reported over the full degree axis in every design in which they were run, and both prove nearly invariant to the nonlinear nuisance axis, which makes them the fixed reference against which the DiD-BCF degree sensitivity is read. The partial unit-intercept route has a small surface-RMSE advantage over the no-unit-intercept implementation in the B2 $d = 1$ audit, but it has not yet been run over the remaining DGPs or degrees.

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1 Scope, result files, and coverage

The report distinguishes the complete current structured grid from the newer route-comparison file. The complete grid is read from

```
DiD_BCF/summaries_B1_*.csv, DiD_BCF/summaries_B2_*.csv,
DiD_BCF/summaries_D_*.csv.
```

These are tidy per-replication summaries with both `plain` and `corrected` methods and the single specification field `structured`. The summary files cover every listed B1 and D design at $d = 1, 2, 3$, and both B2 sweeps at every panel size $N \in \{50, 100, 200, 400, 800\}$ and degree $d \in \{1, 2, 3\}$. The absence of a file is treated as missing coverage, not as a zero or as a completed sweep.

The route-comparison file is

```
Results/identification/routes_raw_B2_sweep_lin_1_full.csv
Results/identification/routes_summary_B2_sweep_lin_1_full.csv
```

It contains several identification routes, but this report retains only

```
structured    structured_rfx_unit
```

The first is the two-way-restricted DiD-BCF fit without a unit random intercept. The second is the same fit with an additive unit-level random intercept. The route file has 15 replications per method, panel size, and method summary. It is not pooled with the 100-replication full-grid summaries.

The benchmark summaries were rebuilt from the current files in

```
R_code/    Results/_staging/DoubleML_Colab/
Results/_staging/CFFE_Wang_Colab/
```

and the current TWFE summaries in `Results/summaries_twfe_*.csv`. The folder named `CFFE.Wang.Colab` contains the grf-DiD implementation associated with Wang, so it is labelled grf-DiD (Wang) rather than CFFE in the tables. The benchmark availability by scenario and panel size is shown in Table 1; the table was regenerated from the actual files before this report was written.

Table 1: Monte-Carlo replications available per scenario, panel size and estimator (maximum over the three linearity degrees). A dash marks a cell that was not run.

Scenario	N	DiD-BCF (plain)	DiD-BCF (corrected)	TWFE	Callaway-Sant'Anna	Gardner (did2s)	DoubleML DR-DiD	Synthetic DiD	grf-DiD (Wang)
B1_baseline	200	100	100	100	100	100	100	100	100
B1_strong_confounder	200	100	100	100	100	100	100	100	100
B1_serial_corr	200	100	100	100	100	100	100	100	100
B1_selection_obs	200	100	100	100	100	100	100	100	100
B1_null	200	100	100	100	200	100	100	100	100
B2_sweep	50	100	100	100	–	100	100	100	100
B2_sweep	100	100	100	100	–	100	100	100	100
B2_sweep	200	100	100	100	100	100	100	100	100
B2_sweep	400	100	100	100	100	100	100	100	100
B2_sweep	800	100	100	100	100	100	100	100	100
B2_sweep	1600	–	–	100	–	–	100	–	100
B2_sweep_serial	50	100	100	100	–	100	100	100	100
B2_sweep_serial	100	100	100	100	–	100	100	100	100
B2_sweep_serial	200	100	100	100	100	100	100	100	100
B2_sweep_serial	400	100	100	100	100	100	100	100	100
B2_sweep_serial	800	100	100	100	100	100	100	100	100
B2_sweep_serial	1600	–	–	100	–	–	100	–	100
D_staggered	200	100	100	100	100	100	100	100	100
D_contamination	200	100	100	100	100	100	100	100	100

Note. Every scenario is additionally run at each linearity degree $d \in \{1, 2, 3\}$. The DoubleML DR-DiD sweep cells are complete at all 5 panel sizes; TWFE is not reported for the staggered scenarios at the cohort-by-time level because pooling cohorts is precisely the contamination it suffers from.

Two benchmark estimators are reported over the full degree axis in every design in which they were run. DoubleML DR-DiD covers all five B1 designs, both D designs, and all five panel sizes of both B2 sweeps, at $d = 1, 2, 3$. grf-DiD (Wang) covers all five B1 designs, both D designs, and all five panel sizes of both B2 sweeps, at $d = 1, 2, 3$. Because both estimators report cohort-time effects rather than a pooled ATT or an event-study coefficient, they appear in the GATT tables rather than in the ATT table, and in the staggered designs they are read cell by cell.

The full grid does not yet contain a completed B1 selection-on-both result, although that design is defined in the DGP code and is described below. It is therefore not silently included as if it had been run. Likewise, the unit random-intercept route is not presented for B1, B2 serial correlation, or D, because no corresponding route summaries are currently present.

2 Mathematical simulation design

2.1 Persistent latent covariates and the degree axis

Each replication draws persistent unit-level covariates once and carries them through the $T = 8$ periods. For unit $i = 1, \dots, N$,

$$X_{1i} \sim \text{Bernoulli}(0.5), \quad (X_{2i}, X_{3i}, X_{4i}) \sim \mathcal{N}(0, 1)^3, \quad X_{5i} \sim \text{Uniform}(-1, 1), \quad (1)$$

with independent draws across units. At $d = 1$ the estimator receives the latent covariates. At $d = 2$ and $d = 3$ the DGP remains a function of the same latent variables, but the estimator receives

$$X_{1i}^o = X_{1i}, \quad X_{2i}^o = X_{2i}, \quad (2)$$

$$X_{3i}^o = \text{std} \left[\left(\frac{X_{3i}X_{4i}}{5} + 0.6 \right)^3 \right], \quad X_{4i}^o = \text{std}[\exp(X_{4i}/2)], \quad (3)$$

$$X_{5i}^o = \text{std}[(X_{3i} + X_{5i} + 2)^2]. \quad (4)$$

Here standardisation is within replication. Thus $d = 2$ changes the parameterisation of the nuisance functions without changing the true outcome or estimands. At $d = 3$, the treatment-effect surface is also changed from an additive linear function to a centered quadratic and interaction while its mean and variance are held fixed.

2.2 Canonical B1 and B2 designs

Let $A_i \sim \mathcal{N}(0, 1)$ and let the unobserved time-invariant unit effect be

$$\alpha_i = \sigma_\alpha A_i. \quad (5)$$

The assignment utility is

$$U_i = \begin{cases} 0.8X_{1i} + 0.6X_{4i}, & \text{selection on observables,} \\ c_\alpha A_i, & \text{selection on the unobservable,} \\ 0.8X_{1i} + 0.6X_{4i} + c_\alpha A_i, & \text{selection on both.} \end{cases} \quad (6)$$

The utility is centered at the target treated share $p_T = 0.5$, and treatment is drawn from

$$T_i \mid X_i, A_i \sim \text{Bernoulli}[\text{logit}^{-1}(1.5[U_i - Q_{1-p_T}(U)] + \nu_i)], \quad \nu_i \sim \mathcal{N}(0, 0.5^2). \quad (7)$$

For the canonical designs, treated units adopt at $G_i = 4$ and never-treated units have $G_i = \infty$. With $t = 0, \dots, 7$,

$$D_{it} = \mathbb{1}\{G_i \leq t\}, \quad k_{it} = t - G_i. \quad (8)$$

The untreated potential outcome is

$$Y_{it}(0) = \beta_0 + \alpha_i + \gamma_t + f(X_i) + s(X_i)t + \varepsilon_{it}, \quad (9)$$

where

$$\beta_0 = -0.5, \quad \gamma_t = 0.2t, \quad (10)$$

$$f(X_i) = -0.75X_{1i} + 0.5X_{2i} - 0.5X_{3i} - 1.3X_{4i} + 1.8X_{5i}, \quad (11)$$

$$s(X_i) = 0.3 \left(0.7X_{4i} + \sqrt{0.51}X_{3i} \right). \quad (12)$$

The observed outcome is

$$Y_{it} = Y_{it}(0) + D_{it}\tau_i. \quad (13)$$

For $d \in \{1, 2\}$, the heterogeneous effect is

$$\tau_i = 3 + 1.5X_{1i} + 0.75X_{2i}. \quad (14)$$

For $d = 3$, it is

$$\tau_i = 3 + 1.5X_{1i} + 0.75 \left[\frac{1}{\sqrt{2}} \left\{ \frac{X_{2i}^2 - 1}{\sqrt{2}} + (2X_{1i} - 1)X_{2i} \right\} \right]. \quad (15)$$

The null design sets $\tau_i = 0$ for every unit. Serial-correlation designs replace iid errors by the stationary AR(1) process

$$\varepsilon_{it} = \rho\varepsilon_{i,t-1} + \sigma_\varepsilon\sqrt{1 - \rho^2} Z_{it}, \quad Z_{it} \sim \mathcal{N}(0, 1). \quad (16)$$

The canonical scenario settings used by the current summaries are

Scenario	σ_α	c_α	Selection	ρ	Effect
B1_baseline	1	1	unobservable	0	heterogeneous
B1_strong_confounder	2	1.5	unobservable	0	heterogeneous
B1_serial_corr	1	1	unobservable	0.6	heterogeneous
B1_selection_obs	1	0	observable	0	heterogeneous
B1_null	1	1	unobservable	0	zero
B2_sweep	1	1	unobservable	0	heterogeneous
B2_sweep_serial	1	1	unobservable	0.6	heterogeneous

The B1 designs use $N = 200$. Both B2 sweeps contain structured DiD-BCF cells at $N \in \{50, 100, 200, 400, 800\}$. The DoubleML and Wang benchmark files now cover the same five panel sizes, and additionally retain a stale $N = 1600$ column from before that size was dropped; the remaining benchmark availability is shown in Table 1.

2.3 Staggered D designs

The D designs assign units to adoption cohorts $G_i \in \{4, 5, 6, \infty\}$, with shares $(0.25, 0.25, 0.25, 0.25)$. Cohort assignment uses a multinomial Gumbel draw. Let s_0 denote the never-treated share, let s_j denote the share of treated cohort j , and let q_i be the selected observable, unobservable, or combined assignment driver. The utilities are

$$V_{i0} = \log(s_0) + \xi_{i0}, \quad (17)$$

$$V_{ij} = \log(s_j) + \left[0.8 + 0.5 \left(1 - \frac{j-1}{K-1} \right) \right] q_i + \xi_{ij}, \quad \xi_{ij} \sim \text{Gumbel}(0, 1), \quad (18)$$

and the selected cohort maximizes V_{ij} . For cohort g_j and event time $k = t - g_j \geq 0$, the effect is

$$\tau_{it}(g_j, k) = M_j(1 + rk)\tau_i, \quad (19)$$

so that

$$Y_{it} = Y_{it}(0) + D_{it}\tau_{it}(G_i, k_{it}). \quad (20)$$

The headline staggered design uses $(M_1, M_2, M_3) = (1, 1.5, 2)$ and $r = 0.4$. The contamination design uses $(M_1, M_2, M_3) = (1, 2, 3)$ and $r = 0.8$. These two designs deliberately create cohort and event-time heterogeneity, the setting in which already-treated-as-control comparisons contaminate pooled TWFE.

3 Estimands and estimators

3.1 True estimands

The individual treatment effect on a treated post-treatment observation is

$$\text{CATT}_{it} = D_{it}\tau_{it}, \quad (21)$$

where $\tau_{it} = \tau_i$ in the canonical designs and Equation (19) applies in the D designs. For a cohort g and calendar time t ,

$$\text{GATT}(g, t) = \frac{1}{n_{g,t}} \sum_{i:G_i=g} \text{CATT}_{it}. \quad (22)$$

The event-study estimand is

$$\text{ES}(k) = \frac{1}{n_k} \sum_{i:t:G_i < \infty, t-G_i=k} \text{CATT}_{it}, \quad (23)$$

and the overall treated-post ATT is

$$\text{ATT} = \frac{1}{n_1} \sum_{i:t:D_{it}=1} \text{CATT}_{it}. \quad (24)$$

In the single-cohort designs, $\text{GATT}(4, 4)$, $\text{ES}(0)$, and the overall ATT target the same post-treatment average, although the estimators report them through different interfaces. In the D designs, cohort and event time matter, so the GATT and event-study results must be read cell by cell.

3.2 DiD-BCF specifications

Both structured implementations are reported as DiD-BCF. The common model is

$$Y_{it} = a(G_i, X_i) + b(t, X_i) + \{\tau_0 + \tau(X_i, k_{it}, G_i)\} D_{it} + u_i + \varepsilon_{it}. \quad (25)$$

For the structured implementation without a unit intercept, $u_i = 0$. For structured DiD-BCF with a unit random intercept, u_i is an additive, time-invariant random effect indexed by unit. The level forest a receives the cohort label and covariates but no time column. The trend forest b receives calendar time and covariates but no cohort label. The treatment forest receives the effect modifiers, event time, and cohort when cohort-specific effects are needed. This is the structural two-way restriction

$$\mu(G_i, t, X_i) = a(G_i, X_i) + b(t, X_i), \quad (26)$$

which prevents the prognostic component from representing a group-by-time treatment interaction. The unit intercept cancels in a time difference and therefore does not reopen that interaction; it additionally absorbs persistent unit-level outcome heterogeneity.

The plain DiD-BCF estimand is the posterior treatment forest evaluated on treated post-treatment observations. The corrected estimand uses the fitted untreated change. For cohort g compared with never-treated units, define

$$\delta_i = \mathbb{1}\{G_i = g\}, \quad \Delta Y_i = Y_{it} - Y_{i,g-1}, \quad m_i^s = \mu_i^s(t) - \mu_i^s(g-1). \quad (27)$$

Let $\hat{\pi}_i$ estimate $P(\delta_i = 1 \mid X_i)$ and let

$$\hat{\gamma}_i = \frac{\delta_i - \hat{\pi}_i}{(1 - \hat{\pi}_i)\bar{\pi}}, \quad A_i = \delta_i - (1 - \delta_i) \frac{\hat{\pi}_i}{1 - \hat{\pi}_i}. \quad (28)$$

With Bayesian-bootstrap weights W_i^s , the corrected draw is

$$\hat{\theta}_{g,t}^s = \frac{\sum_i W_i^s A_i (\Delta Y_i - m_i^s)}{\sum_i W_i^s \delta_i}, \quad (29)$$

$$\hat{b}_{g,t}^s = \frac{1}{n_g} \sum_i \hat{\gamma}_i (\bar{m}_i - m_i^s), \quad (30)$$

$$\tilde{\theta}_{g,t}^s = \hat{\theta}_{g,t}^s - \hat{b}_{g,t}^s, \quad \bar{m}_i = \frac{1}{S} \sum_{s=1}^S m_i^s. \quad (31)$$

The corrected point estimate and interval are the posterior mean and empirical quantiles of the corrected draws. Aggregated ATT and event-study draws use treated-count weights over the relevant cohort-time cells.

3.3 Benchmark estimators

The comparison includes TWFE, Callaway–Sant’Anna’s doubly robust group-time estimator, Gardner’s two-stage event-study estimator, Synthetic DiD, the DoubleML doubly robust DiD estimator, and grf-DiD (Wang). The GATT-only estimators are broadcast to treated observations when the individual CATT surface is evaluated. This is a deliberately transparent benchmark: it gives cell-level estimators their natural scalar prediction while preserving the distinction between scalar identification and individual heterogeneity.

4 Evaluation formulas

Let $\hat{\theta}_r$ be the estimate in replication r , let θ_r be the replication-specific realised target, and let $e_r = \hat{\theta}_r - \theta_r$. The main scalar metrics are

$$\text{Bias} = \frac{1}{R} \sum_{r=1}^R e_r, \quad (32)$$

$$\text{RMSE} = \left(\frac{1}{R} \sum_{r=1}^R e_r^2 \right)^{1/2}, \quad (33)$$

$$\text{MAE} = \frac{1}{R} \sum_{r=1}^R |e_r|, \quad (34)$$

$$\text{MAPE} = \frac{1}{|\mathcal{R}_+|} \sum_{r \in \mathcal{R}_+} \left| \frac{e_r}{\theta_r} \right|, \quad \mathcal{R}_+ = \{r : |\theta_r| > 10^{-8}\}. \quad (35)$$

Because the target itself varies across replications, the tables also report the Monte-Carlo standard deviation of the estimation error,

$$\text{SD}(e) = \left[\frac{1}{R-1} \sum_{r=1}^R (e_r - \bar{e})^2 \right]^{1/2}, \quad (36)$$

$$\text{SD}_{\text{table}}(e) = [\text{RMSE}^2 - \text{Bias}^2]^{1/2}. \quad (37)$$

For an interval $[L_r, U_r]$,

$$\text{Coverage}_{1-\alpha} = \frac{1}{R} \sum_{r=1}^R \mathbb{1}\{L_r \leq \theta_r \leq U_r\}, \quad (38)$$

$$\text{Length}_{1-\alpha} = \frac{1}{R} \sum_{r=1}^R (U_r - L_r), \quad (39)$$

$$\text{SDRatio} = \frac{R^{-1} \sum_{r=1}^R s_r}{\text{SD}(\hat{\theta})}. \quad (40)$$

Here s_r is the reported posterior standard deviation or standard error. The Monte-Carlo standard errors of the bias and coverage are

$$\text{MCSE}(\text{Bias}) = \frac{\text{SD}(\hat{\theta})}{\sqrt{R}}, \quad \text{MCSE}(\text{Coverage}) = \left[\frac{\hat{p}(1-\hat{p})}{R} \right]^{1/2}. \quad (41)$$

For the individual effect surface, let $\widehat{\text{CATT}}_{ir}$ be the prediction for treated observation i in replication r and let n_r^{tr} be the number of treated post-treatment observations. The within-replication surface metrics are

$$\text{RMSE}_{\text{surf},r} = \left[\frac{1}{n_r^{\text{tr}}} \sum_{i:D_{ir}=1} (\widehat{\text{CATT}}_{ir} - \text{CATT}_{ir})^2 \right]^{1/2}, \quad (42)$$

$$\text{MAE}_{\text{surf},r} = \frac{1}{n_r^{\text{tr}}} \sum_{i:D_{ir}=1} |\widehat{\text{CATT}}_{ir} - \text{CATT}_{ir}|, \quad (43)$$

$$\text{MAPE}_{\text{surf},r} = \frac{1}{|\mathcal{I}_r|} \sum_{i \in \mathcal{I}_r} \left| \frac{\widehat{\text{CATT}}_{ir} - \text{CATT}_{ir}}{\text{CATT}_{ir}} \right|, \quad \mathcal{I}_r = \{i : D_{ir} = 1, |\text{CATT}_{ir}| > 10^{-8}\}. \quad (44)$$

The reported surface values average these within-replication quantities over replications. Finally, the finite-sample rate diagnostic is

$$Z_r = \sqrt{N}(\hat{\theta}_r - \theta_r), \quad (45)$$

whose mean should approach zero and whose standard deviation should stabilize for a valid $\sqrt{N}$ approximation.

5 Coverage and complete-grid results

5.1 B1 scalar estimands

The decomposed ATT results are reported in Table 2, and the directly comparable GATT(4,4) results are in Table 3. DoubleML DR-DiD and grf-DiD do not appear in the ATT table because neither reports a pooled ATT; both report cohort-time effects, so their canonical-design results are in Table 3, at all three degrees. At $d = 1$, the structured DiD-BCF point estimates are close to the true target in the baseline, strong-confounder, serial-correlation, and selection-on-observables designs. The plain specification is especially stable in the baseline and strong-confounder cases. The corrected specification generally has a larger posterior spread, which is useful under serial correlation but can become costly under transformed covariates.

The degree axis is now informative. The $d = 2$ and $d = 3$ plain structured estimates remain close to zero in the canonical scalar comparisons, although their uncertainty calibration deteriorates in the serial and selection-on-observables settings. The corrected summaries show

much larger RMSE and reported-spread distortions at $d = 2$ and $d = 3$, especially under selection-on-observables. This is evidence about the current posterior correction implementation and sampler budget, not evidence that the DGP has become unidentified: the DGP truth is unchanged at $d = 2$, and the $d = 3$ effect surface is explicitly known.

The selection-on-observables case is a useful stress test. The structured DiD-BCF plain estimator remains close to the benchmark point estimates at all three degrees, while Synthetic DiD has a visibly larger ATT bias in the current files. The result supports the value of conditioning on the covariates, but its uncertainty advantage should not be claimed until the nonlinear corrected cells receive more replications and a longer sampling budget.

Table 2: Decomposed Monte-Carlo metrics for the overall ATT on the canonical-DiD scenarios.

Estimator	d	Bias	SD(err)	RMSE	Cov.90	Cov.95	Len.95	$\overline{\text{sd}}/\text{SD}$
<i>B1_baseline</i>								
DiD-BCF (plain)	1	0.000	0.098	0.098	0.920	0.960	0.425	0.839
DiD-BCF (corrected)	1	-0.000	0.179	0.179	0.930	0.980	0.813	1.116
TWFE	1	-0.024	0.177	0.179	0.970	0.980	0.837	1.067
Synthetic DiD	1	-0.023	0.168	0.169	0.970	1.000	0.797	1.064
DiD-BCF (plain)	2	-0.003	0.115	0.115	0.890	0.940	0.437	0.796
DiD-BCF (corrected)	2	0.116	1.064	1.070	0.940	0.970	1.502	0.359
TWFE	2	-0.024	0.177	0.179	0.970	0.980	0.837	1.067
Synthetic DiD	2	-0.023	0.168	0.169	0.970	1.000	0.797	1.064
DiD-BCF (plain)	3	-0.010	0.115	0.115	0.900	0.910	0.444	0.700
DiD-BCF (corrected)	3	0.112	1.091	1.096	0.930	0.970	1.520	0.354
TWFE	3	-0.024	0.177	0.179	0.970	0.980	0.837	1.067
Synthetic DiD	3	-0.023	0.168	0.169	0.960	1.000	0.797	1.023
<i>B1_strong_confounder</i>								
DiD-BCF (plain)	1	-0.000	0.096	0.096	0.940	0.980	0.425	0.868
DiD-BCF (corrected)	1	0.008	0.176	0.176	0.950	0.990	0.819	1.150
TWFE	1	-0.022	0.179	0.180	0.970	0.980	0.837	1.097
Synthetic DiD	1	-0.027	0.169	0.171	0.940	0.990	0.795	1.086
DiD-BCF (plain)	2	-0.005	0.108	0.108	0.910	0.940	0.441	0.842
DiD-BCF (corrected)	2	0.037	0.357	0.359	0.940	0.960	1.112	0.774
TWFE	2	-0.022	0.179	0.180	0.970	0.980	0.837	1.097
Synthetic DiD	2	-0.027	0.169	0.171	0.940	0.990	0.795	1.086
DiD-BCF (plain)	3	-0.011	0.104	0.105	0.900	0.940	0.445	0.741
DiD-BCF (corrected)	3	0.033	0.359	0.361	0.930	0.970	1.114	0.765
TWFE	3	-0.022	0.179	0.180	0.970	0.980	0.837	1.097
Synthetic DiD	3	-0.027	0.169	0.171	0.940	0.990	0.796	1.035
<i>B1_serial_corr</i>								
DiD-BCF (plain)	1	0.005	0.126	0.126	0.800	0.830	0.360	0.596
DiD-BCF (corrected)	1	-0.002	0.150	0.150	0.930	0.980	0.704	1.096
TWFE	1	-0.022	0.197	0.198	0.950	0.980	0.894	1.038
Synthetic DiD	1	-0.024	0.164	0.166	0.940	0.990	0.792	1.081
DiD-BCF (plain)	2	0.000	0.143	0.143	0.700	0.820	0.378	0.587
DiD-BCF (corrected)	2	0.071	0.475	0.480	0.930	0.960	1.234	0.636
TWFE	2	-0.022	0.197	0.198	0.950	0.980	0.894	1.038
Synthetic DiD	2	-0.024	0.164	0.166	0.940	0.990	0.792	1.081
DiD-BCF (plain)	3	-0.010	0.137	0.137	0.740	0.860	0.382	0.562
DiD-BCF (corrected)	3	0.078	0.596	0.601	0.940	0.960	1.268	0.538
TWFE	3	-0.022	0.197	0.198	0.950	0.980	0.894	1.038
Synthetic DiD	3	-0.024	0.164	0.166	0.940	0.990	0.793	1.038
<i>B1_selection_obs</i>								
DiD-BCF (plain)	1	0.056	0.111	0.124	0.900	0.930	0.467	0.757
DiD-BCF (corrected)	1	0.002	0.244	0.244	0.920	0.970	1.012	0.984
TWFE	1	-0.023	0.223	0.224	0.890	0.950	0.833	0.868
Synthetic DiD	1	0.417	0.183	0.455	0.320	0.410	0.778	0.930

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Table 2 continued from the previous page.

Estimator	d	Bias	SD(err)	RMSE	Cov.90	Cov.95	Len.95	$\overline{\text{sd}}/\text{SD}$
DiD-BCF (plain)	2	0.053	0.139	0.149	0.800	0.880	0.482	0.698
DiD-BCF (corrected)	2	0.165	1.328	1.338	0.910	0.970	2.407	0.467
TWFE	2	-0.023	0.223	0.224	0.890	0.950	0.833	0.868
Synthetic DiD	2	0.417	0.183	0.455	0.320	0.410	0.778	0.930
DiD-BCF (plain)	3	0.046	0.140	0.148	0.810	0.900	0.494	0.701
DiD-BCF (corrected)	3	0.163	1.352	1.361	0.910	0.980	2.371	0.465
TWFE	3	-0.023	0.223	0.224	0.890	0.950	0.833	0.868
Synthetic DiD	3	0.417	0.183	0.455	0.320	0.420	0.780	0.999

Note. $N = 200$, 100 replications, d is the linearity degree. Bias and RMSE are Monte-Carlo moments of the estimation error and SD(err) is its standard deviation, $\sqrt{\text{RMSE}^2 - \text{Bias}^2}$ (the target itself varies by replication, so the raw dispersion of the estimate is not the error SD); Cov.90/95 are credible/confidence-interval coverage rates; Len.95 the average 95% interval length; $\overline{\text{sd}}/\text{SD}$ the calibration ratio of the average reported posterior/standard error to the Monte-Carlo SD (1 is perfect calibration, < 1 over-confident).

Table 3: Decomposed Monte-Carlo metrics for GATT($g=4, t=4$) on the canonical-DiD scenarios – the cell every estimator reports.

Estimator	d	Bias	SD(err)	RMSE	Cov.90	Cov.95	Len.95	$\overline{\text{sd}}/\text{SD}$
<i>B1_baseline</i>								
DiD-BCF (plain)	1	-0.022	0.108	0.110	0.970	1.000	0.558	1.005
DiD-BCF (corrected)	1	-0.028	0.229	0.231	0.900	0.950	0.979	1.047
Callaway–Sant’Anna	1	-0.031	0.219	0.221	0.870	0.970	0.902	1.031
DoubleML DR-DiD	1	-0.039	0.301	0.304	0.870	0.920	1.157	0.941
grf-DiD (Wang)	1	-0.036	0.219	0.222	0.730	0.810	0.630	0.731
DiD-BCF (plain)	2	-0.024	0.119	0.122	0.950	0.980	0.572	0.985
DiD-BCF (corrected)	2	0.109	1.034	1.040	0.870	0.960	1.676	0.411
Callaway–Sant’Anna	2	-0.034	0.223	0.225	0.850	0.960	0.914	1.020
DoubleML DR-DiD	2	-0.034	0.278	0.280	0.920	0.940	1.127	1.020
grf-DiD (Wang)	2	-0.037	0.220	0.223	0.710	0.810	0.630	0.730
DiD-BCF (plain)	3	-0.031	0.120	0.124	0.940	0.980	0.577	0.890
DiD-BCF (corrected)	3	0.094	1.025	1.029	0.880	0.970	1.659	0.414
Callaway–Sant’Anna	3	-0.034	0.223	0.225	0.860	0.960	0.908	0.936
DoubleML DR-DiD	3	-0.038	0.268	0.271	0.890	0.970	1.115	1.044
grf-DiD (Wang)	3	-0.037	0.220	0.223	0.720	0.810	0.630	0.729
<i>B1_strong_confounder</i>								
DiD-BCF (plain)	1	-0.012	0.101	0.101	0.990	1.000	0.558	1.050
DiD-BCF (corrected)	1	-0.011	0.226	0.226	0.890	0.980	0.985	1.065
Callaway–Sant’Anna	1	-0.017	0.203	0.204	0.940	0.980	0.904	1.076
DoubleML DR-DiD	1	-0.016	0.280	0.281	0.930	0.960	1.138	1.020
grf-DiD (Wang)	1	-0.018	0.201	0.201	0.790	0.860	0.629	0.782
DiD-BCF (plain)	2	-0.016	0.110	0.111	0.960	0.990	0.572	1.017
DiD-BCF (corrected)	2	0.004	0.412	0.412	0.910	0.960	1.282	0.777
Callaway–Sant’Anna	2	-0.021	0.208	0.209	0.910	0.980	0.904	1.049
DoubleML DR-DiD	2	-0.032	0.251	0.253	0.950	0.990	1.133	1.112
grf-DiD (Wang)	2	-0.018	0.200	0.201	0.780	0.860	0.629	0.783
DiD-BCF (plain)	3	-0.021	0.105	0.107	0.950	1.000	0.577	0.965
DiD-BCF (corrected)	3	-0.001	0.407	0.407	0.910	0.970	1.283	0.783
Callaway–Sant’Anna	3	-0.021	0.208	0.209	0.920	0.970	0.912	0.996
DoubleML DR-DiD	3	-0.023	0.255	0.256	0.900	0.960	1.142	1.085
grf-DiD (Wang)	3	-0.018	0.200	0.201	0.790	0.860	0.629	0.782
<i>B1_serial_corr</i>								
DiD-BCF (plain)	1	-0.012	0.111	0.111	0.910	0.970	0.462	0.774
DiD-BCF (corrected)	1	-0.024	0.134	0.136	0.940	0.990	0.693	1.098
Callaway–Sant’Anna	1	-0.025	0.126	0.129	0.970	0.990	0.653	1.098
DoubleML DR-DiD	1	-0.029	0.163	0.165	0.960	0.990	0.776	1.087

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Table 3 continued from the previous page.

Estimator	d	Bias	SD(err)	RMSE	Cov.90	Cov.95	Len.95	$\overline{\text{sd}}/\text{SD}$
grf-DiD (Wang)	1	-0.032	0.130	0.133	0.840	0.910	0.458	0.809
DiD-BCF (plain)	2	-0.013	0.127	0.128	0.860	0.960	0.478	0.763
DiD-BCF (corrected)	2	0.019	0.268	0.269	0.970	0.990	1.160	1.065
Callaway-Sant'Anna	2	-0.028	0.132	0.135	0.950	0.990	0.662	1.067
DoubleML DR-DiD	2	-0.022	0.169	0.170	0.930	0.970	0.773	1.102
grf-DiD (Wang)	2	-0.032	0.130	0.133	0.840	0.900	0.458	0.809
DiD-BCF (plain)	3	-0.025	0.120	0.122	0.900	0.960	0.481	0.754
DiD-BCF (corrected)	3	-0.011	0.155	0.156	0.990	0.990	1.163	1.626
Callaway-Sant'Anna	3	-0.028	0.132	0.135	0.960	0.980	0.664	0.989
DoubleML DR-DiD	3	-0.024	0.169	0.170	0.940	0.980	0.772	1.008
grf-DiD (Wang)	3	-0.032	0.129	0.133	0.840	0.910	0.458	0.811
<i>B1_selection_obs</i>								
DiD-BCF (plain)	1	0.026	0.124	0.127	0.950	0.950	0.598	0.869
DiD-BCF (corrected)	1	-0.003	0.317	0.317	0.930	0.970	1.256	0.946
Callaway-Sant'Anna	1	-0.006	0.280	0.280	0.900	0.950	1.082	0.910
DoubleML DR-DiD	1	-0.011	0.343	0.344	0.890	0.940	1.264	0.917
grf-DiD (Wang)	1	-0.099	0.233	0.254	0.670	0.790	0.615	0.650
DiD-BCF (plain)	2	0.025	0.140	0.142	0.930	0.960	0.617	0.840
DiD-BCF (corrected)	2	0.232	1.714	1.729	0.910	0.970	2.861	0.432
Callaway-Sant'Anna	2	0.018	0.336	0.337	0.888	0.910	1.153	0.827
DoubleML DR-DiD	2	-0.016	0.336	0.336	0.880	0.940	1.251	0.925
grf-DiD (Wang)	2	-0.100	0.233	0.253	0.680	0.790	0.615	0.651
DiD-BCF (plain)	3	0.015	0.140	0.141	0.930	0.950	0.628	0.885
DiD-BCF (corrected)	3	0.233	1.715	1.731	0.930	0.970	2.867	0.438
Callaway-Sant'Anna	3	0.018	0.336	0.337	0.888	0.921	1.155	0.855
DoubleML DR-DiD	3	-0.036	0.337	0.339	0.900	0.920	1.294	0.951
grf-DiD (Wang)	3	-0.099	0.234	0.254	0.670	0.800	0.615	0.649

Note. $N = 200$, 100 replications, d is the linearity degree. Bias and RMSE are Monte-Carlo moments of the estimation error and SD(err) is its standard deviation, $\sqrt{\text{RMSE}^2 - \text{Bias}^2}$ (the target itself varies by replication, so the raw dispersion of the estimate is not the error SD); Cov.90/95 are credible/confidence-interval coverage rates; Len.95 the average 95% interval length; $\overline{\text{sd}}/\text{SD}$ the calibration ratio of the average reported posterior/standard error to the Monte-Carlo SD (1 is perfect calibration, < 1 over-confident).

5.2 Sharp null

The sharp-null results in Table 4 are considerably better than the pre-correction outputs. The plain structured DiD-BCF estimates are centered near zero and its two-sided rejection rates are 0.01 to 0.05 across the ATT, event-study, and cohort-time null cells: the test is conservative rather than oversized, which is the safe direction and is the behaviour the conservative-under-the-null argument requires. The corrected method rejects in 0.03 to 0.09 of replications. With 100 replications the Monte-Carlo standard error of a rejection rate near nominal is about 0.022, so every corrected cell lies within roughly two standard errors of the nominal five percent and none of them supports a claim of size distortion.

The informative null comparison is with the benchmarks. DoubleML DR-DiD is essentially exact, at 0.03 to 0.07. Callaway-Sant'Anna is mildly oversized at 0.03 to 0.11. grf-DiD (Wang) is the clear outlier, rejecting a true null in 0.11 to 0.25 of replications with 95 percent coverage as low as 0.75, so its individual-effect estimates come with intervals that are materially too narrow.

The remaining null concern is dispersion, not size. The corrected error SD at $d = 2$ and $d = 3$ rises to roughly 1.0 against 0.11 for the plain fit, even though the true effect is identically zero. That is the same nonlinear-degree instability documented in the B2 and D sections, and it is a precision problem in the correction rather than a failure of the null calibration.

Table 4: Behaviour under the sharp null $\tau \equiv 0$ (scenario **B1.null**): bias, dispersion, coverage and the empirical size of the two-sided 5% test.

Estimator	d	Bias	SD(err)	Cov.90	Cov.95	Size (5%)	MCSE
<i>ATT, ATT</i>							
DiD-BCF (plain)	1	0.003	0.097	0.930	0.960	0.040	0.020
DiD-BCF (corrected)	1	-0.001	0.181	0.890	0.940	0.060	0.024
TWFE	1	-0.022	0.193	0.930	0.955	0.045	0.015
Synthetic DiD	1	-0.023	0.168	0.880	0.960	0.040	0.020
DiD-BCF (plain)	2	-0.004	0.113	0.880	0.950	0.050	0.022
DiD-BCF (corrected)	2	0.123	1.087	0.860	0.960	0.040	0.020
TWFE	2	-0.022	0.193	0.930	0.955	0.045	0.015
Synthetic DiD	2	-0.023	0.168	0.880	0.960	0.040	0.020
DiD-BCF (plain)	3	-0.004	0.113	0.880	0.950	0.050	0.022
DiD-BCF (corrected)	3	0.123	1.087	0.860	0.960	0.040	0.020
TWFE	3	-0.022	0.193	0.930	0.955	0.045	0.015
Synthetic DiD	3	-0.023	0.168	0.880	0.960	0.040	0.020
<i>ES, k=0</i>							
DiD-BCF (plain)	1	-0.018	0.107	0.960	0.990	0.010	0.010
DiD-BCF (corrected)	1	-0.030	0.233	0.830	0.910	0.090	0.029
TWFE	1	-0.022	0.185	0.905	0.970	0.030	0.012
Gardner (did2s)	1	-0.047	0.173	0.940	0.970	0.030	0.017
DiD-BCF (plain)	2	-0.022	0.116	0.950	0.990	0.010	0.010
DiD-BCF (corrected)	2	0.103	0.965	0.860	0.920	0.080	0.027
TWFE	2	-0.022	0.185	0.905	0.970	0.030	0.012
Gardner (did2s)	2	-0.047	0.173	0.940	0.970	0.030	0.017
DiD-BCF (plain)	3	-0.022	0.116	0.950	0.990	0.010	0.010
DiD-BCF (corrected)	3	0.103	0.965	0.860	0.920	0.080	0.027
TWFE	3	-0.022	0.185	0.905	0.970	0.030	0.012
Gardner (did2s)	3	-0.047	0.173	0.940	0.970	0.030	0.017
<i>GATT, g=4, t=4</i>							
DiD-BCF (plain)	1	-0.018	0.107	0.960	0.990	0.010	0.010
DiD-BCF (corrected)	1	-0.030	0.233	0.830	0.910	0.090	0.029
Callaway-Sant'Anna	1	-0.031	0.219	0.840	0.930	0.070	0.026
DoubleML DR-DiD	1	-0.031	0.281	0.890	0.950	0.050	0.022
grf-DiD (Wang)	1	-0.033	0.211	0.720	0.750	0.250	0.043
DiD-BCF (plain)	2	-0.022	0.116	0.950	0.990	0.010	0.010
DiD-BCF (corrected)	2	0.103	0.965	0.860	0.920	0.080	0.027
Callaway-Sant'Anna	2	-0.034	0.223	0.800	0.910	0.090	0.029
DoubleML DR-DiD	2	-0.054	0.273	0.870	0.950	0.050	0.022
grf-DiD (Wang)	2	-0.033	0.210	0.720	0.750	0.250	0.043
DiD-BCF (plain)	3	-0.022	0.116	0.950	0.990	0.010	0.010
DiD-BCF (corrected)	3	0.103	0.965	0.860	0.920	0.080	0.027
Callaway-Sant'Anna	3	-0.034	0.223	0.810	0.890	0.110	0.031
DoubleML DR-DiD	3	-0.055	0.249	0.920	0.970	0.030	0.017
grf-DiD (Wang)	3	-0.033	0.211	0.720	0.750	0.250	0.043

Note. $N = 200$; 100 replications for the model-based estimators and 200 for TWFE. A nominal 5% test should reject in 0.05 of replications; MCSE is the Monte-Carlo standard error of that rejection rate.

6 B2 sample-size sweeps

6.1 Scalar convergence and serial correlation

Table 5 reports the common GATT(4, 4) target at $d = 1$. The structured DiD-BCF point estimates have small bias at all five panel sizes in both sweeps, and both bias and error dispersion decline with N . In the ordinary sweep, the plain and corrected RMSEs fall from 0.250 and 0.707 at $N = 50$ through 0.110 and 0.231 at the base panel to 0.068 and 0.107 at $N = 800$, with 95

percent coverage between 0.94 and 1.00 throughout. The plain fit has the smallest RMSE of any estimator at every panel size, by a factor of about 1.8 over Callaway–Sant’Anna at $N = 50$ and about 1.6 at $N = 800$. DoubleML follows the same convergence pattern from the largest starting error, 0.729 at $N = 50$ falling to 0.114 at $N = 800$, with coverage between 0.96 and 0.98. Wang has small point bias but remains less well covered, with coverage between 0.81 and 0.94 and a calibration ratio that never exceeds 0.81.

The serial sweep gives a different lesson. The plain structured estimator is undercovered at every panel size, at 0.88, 0.93, 0.97, 0.92 and 0.90, with a calibration ratio between 0.64 and 0.77 that does not improve with N ; on the pooled ATT the same shortfall reads 0.83 coverage at every size. The corrected estimator has near-nominal or above-nominal coverage at every size and small bias, at the cost of a wider interval. Because the shortfall is flat in N it is a missing variance component rather than a small-sample artefact, which makes the serial sweep the strongest current evidence in favor of retaining the correction for dependent within-unit errors. DoubleML remains well calibrated across the sweep, while Wang’s coverage sits between 0.89 and 0.94.

Table 5: Sample-size sweep for GATT($g=4, t=4$) at linearity degree $d = 1$: the estimand every method reports, so all estimators are directly comparable across N .

Estimator	N	Bias	SD(err)	RMSE	Cov.95	Len.95	$\overline{\text{sd}}/\text{SD}$
<i>B2_sweep</i>							
DiD-BCF (plain)	50	-0.019	0.249	0.250	0.970	1.073	0.801
DiD-BCF (plain)	100	0.014	0.168	0.168	0.980	0.772	1.004
DiD-BCF (plain)	200	-0.022	0.108	0.110	1.000	0.558	1.005
DiD-BCF (plain)	400	0.004	0.086	0.086	0.970	0.401	0.959
DiD-BCF (plain)	800	0.002	0.068	0.068	0.970	0.288	0.800
DiD-BCF (corrected)	50	0.083	0.702	0.707	0.940	2.889	0.950
DiD-BCF (corrected)	100	0.019	0.341	0.342	0.950	1.567	1.147
DiD-BCF (corrected)	200	-0.028	0.229	0.231	0.950	0.979	1.047
DiD-BCF (corrected)	400	0.001	0.154	0.154	0.950	0.652	0.995
DiD-BCF (corrected)	800	0.001	0.107	0.107	0.980	0.450	0.912
Callaway–Sant’Anna	50	0.055	0.452	0.455	0.940	1.898	0.905
Callaway–Sant’Anna	100	0.013	0.290	0.290	0.970	1.301	1.091
Callaway–Sant’Anna	200	-0.031	0.219	0.221	0.980	0.905	1.034
Callaway–Sant’Anna	400	0.005	0.148	0.148	0.950	0.631	0.991
Callaway–Sant’Anna	800	0.002	0.107	0.107	0.980	0.445	0.893
DoubleML DR-DiD	50	0.045	0.728	0.729	0.980	3.061	0.974
DoubleML DR-DiD	100	-0.003	0.467	0.467	0.970	1.867	1.008
DoubleML DR-DiD	200	-0.038	0.283	0.285	0.960	1.155	1.015
DoubleML DR-DiD	400	-0.001	0.178	0.178	0.960	0.754	0.989
DoubleML DR-DiD	800	0.005	0.114	0.114	0.970	0.503	0.966
DoubleML DR-DiD	1600	0.014	0.072	0.073	0.990	0.349	1.068
grf-DiD (Wang)	50	0.009	0.409	0.409	0.940	1.424	0.758
grf-DiD (Wang)	100	0.006	0.298	0.298	0.890	0.970	0.809
grf-DiD (Wang)	200	-0.036	0.219	0.222	0.810	0.630	0.731
grf-DiD (Wang)	400	0.005	0.148	0.148	0.890	0.445	0.734
grf-DiD (Wang)	800	0.001	0.111	0.111	0.810	0.319	0.646
grf-DiD (Wang)	1600	0.009	0.067	0.067	0.900	0.228	0.767
<i>B2_sweep_serial</i>							
DiD-BCF (plain)	50	-0.045	0.285	0.288	0.880	0.903	0.644
DiD-BCF (plain)	100	0.008	0.173	0.173	0.930	0.640	0.774
DiD-BCF (plain)	200	-0.012	0.111	0.111	0.970	0.462	0.774
DiD-BCF (plain)	400	0.007	0.093	0.093	0.920	0.329	0.730
DiD-BCF (plain)	800	0.003	0.067	0.067	0.900	0.235	0.668
DiD-BCF (corrected)	50	0.030	0.477	0.478	0.970	1.955	0.911
DiD-BCF (corrected)	100	0.014	0.200	0.201	0.990	1.098	1.259

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Table 5 continued from the previous page.

Estimator	N	Bias	SD(err)	RMSE	Cov.95	Len.95	$\overline{\text{sd}}/\text{SD}$
DiD-BCF (corrected)	200	-0.024	0.134	0.136	0.990	0.693	1.098
DiD-BCF (corrected)	400	0.001	0.090	0.090	1.000	0.470	1.065
DiD-BCF (corrected)	800	0.006	0.071	0.071	0.990	0.328	0.880
Callaway-Sant'Anna	50	0.016	0.300	0.301	0.980	1.361	0.889
Callaway-Sant'Anna	100	0.016	0.176	0.177	0.980	0.940	1.166
Callaway-Sant'Anna	200	-0.025	0.126	0.129	0.990	0.653	1.098
Callaway-Sant'Anna	400	0.004	0.087	0.087	0.990	0.460	1.061
Callaway-Sant'Anna	800	0.005	0.070	0.070	0.990	0.326	0.872
DoubleML DR-DiD	50	0.033	0.497	0.498	0.980	2.290	0.995
DoubleML DR-DiD	100	0.033	0.268	0.270	0.990	1.305	1.145
DoubleML DR-DiD	200	-0.018	0.167	0.168	0.970	0.806	1.081
DoubleML DR-DiD	400	-0.004	0.108	0.108	0.980	0.529	1.077
DoubleML DR-DiD	800	0.006	0.082	0.082	0.950	0.365	0.859
DoubleML DR-DiD	1600	0.011	0.047	0.048	0.990	0.248	0.963
grf-DiD (Wang)	50	-0.005	0.290	0.291	0.940	1.091	0.755
grf-DiD (Wang)	100	0.013	0.185	0.186	0.930	0.731	0.904
grf-DiD (Wang)	200	-0.032	0.130	0.133	0.910	0.458	0.809
grf-DiD (Wang)	400	0.004	0.090	0.090	0.910	0.334	0.832
grf-DiD (Wang)	800	0.007	0.074	0.074	0.890	0.244	0.671
grf-DiD (Wang)	1600	0.008	0.042	0.043	0.970	0.173	0.792

Note. 100 replications per cell. A consistent estimator has $|\text{Bias}| \rightarrow 0$ and $\text{SD} \propto N^{-1/2}$; the $\sqrt{N}$ -scaled version of this table is Figure 1.

The figures below show the ordinary-sweep absolute bias, error dispersion, coverage, and the corresponding $\sqrt{N}$ diagnostics over the complete five panel-size grid.

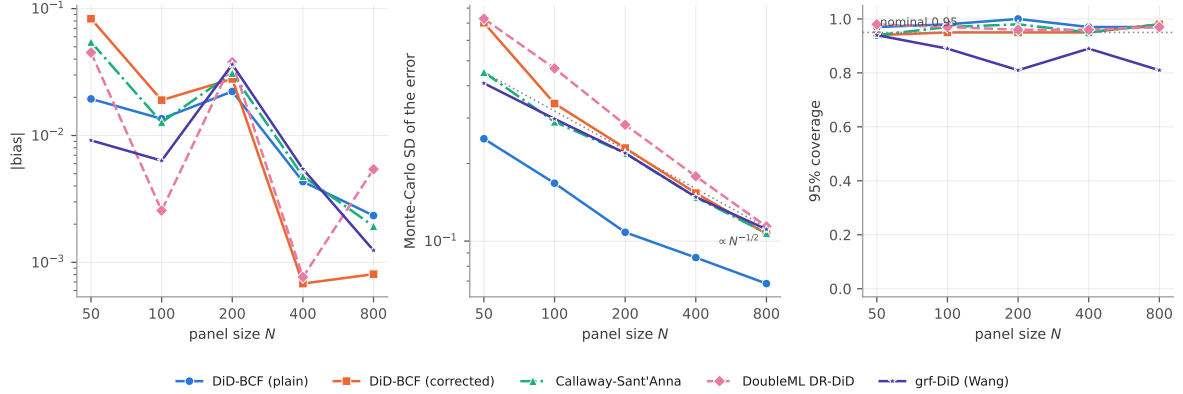


Figure 1: B2 ordinary-sweep bias, error dispersion, and 95% coverage at linearity degree $d = 1$.

6.2 The nonlinear degrees in B2

The nonlinear degrees are reported in Table 6, which carries the same GATT(4,4) target and the same estimators as Table 5. At $d = 2$, the plain scalar results remain close to zero across the ordinary and serial cells, and its calibration ratio stays between 0.80 and 0.99 in the ordinary sweep. The corrected fit does not: its ordinary-sweep calibration ratio reads 0.48, 0.60, 0.41, 0.30 and 0.51 across the five panel sizes, and its RMSE reaches 2.517 at $N = 50$ and 1.319 at $N = 400$ against the plain fit's 0.263 and 0.092. At $d = 3$ the pattern repeats, with a corrected ratio between 0.41 and 0.67 and an RMSE of 2.455 at $N = 50$. The serial cells behave the same way apart from an isolated ratio above one at $N = 200$. Adding the two small panels therefore

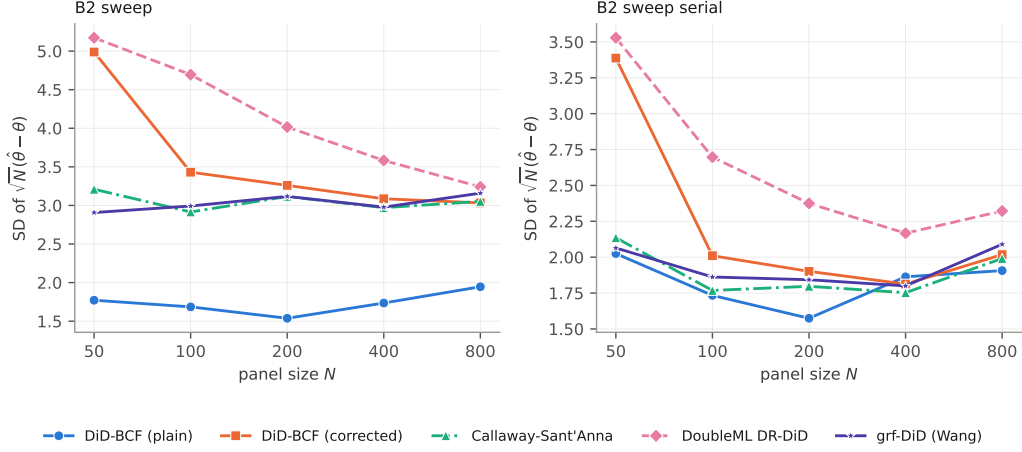


Figure 2: $\sqrt{N}$ -scaled error summaries for the ordinary and serial B2 sweeps at $d = 1$.

sharpens rather than softens the earlier reading: the correction is unstable at the nonlinear degrees across the whole sweep, and the instability is worst where the panel is smallest.

The benchmarks behave very differently along this axis. DoubleML DR-DiD and grf-DiD are nearly invariant to the degree: their bias, dispersion, and interval length at $d = 2$ and $d = 3$ remain close to their $d = 1$ values across all five panel sizes. This makes them useful fixed references against which the DiD-BCF degree sensitivity can be read. Their weaknesses are also unchanged by the degree: grf-DiD remains materially undercovered, at roughly 0.81 to 0.92 in the ordinary sweep and 0.88 to 0.96 in the serial sweep, whereas DoubleML remains well calibrated with wider benchmark intervals.

The point-estimation result is encouraging but the inferential result is not final. The degree axis changes the nuisance problem, so a stable $d = 1$ cell cannot be used as evidence that the $d = 2$ and $d = 3$ corrected intervals are already calibrated.

Table 6: Sample-size sweep for GATT($g=4, t=4$) at the nonlinear degrees $d = 2$ and $d = 3$.

Estimator	N	d	Bias	SD(err)	RMSE	Cov.95	Len.95	$\overline{\text{sd}}/\text{SD}$
<i>B2_sweep</i>								
DiD-BCF (plain)	50	2	-0.024	0.262	0.263	0.960	1.095	0.796
DiD-BCF (plain)	50	3	-0.032	0.275	0.277	0.950	1.109	0.922
DiD-BCF (plain)	100	2	0.010	0.182	0.182	0.980	0.794	0.978
DiD-BCF (plain)	100	3	0.000	0.185	0.185	0.970	0.806	1.066
DiD-BCF (plain)	200	2	-0.024	0.119	0.122	0.980	0.572	0.985
DiD-BCF (plain)	200	3	-0.031	0.120	0.124	0.980	0.577	0.890
DiD-BCF (plain)	400	2	0.012	0.091	0.092	0.960	0.415	0.947
DiD-BCF (plain)	400	3	0.010	0.090	0.091	0.960	0.414	1.029
DiD-BCF (plain)	800	2	-0.001	0.072	0.072	0.960	0.302	0.821
DiD-BCF (plain)	800	3	-0.003	0.071	0.071	0.950	0.299	0.731
DiD-BCF (corrected)	50	2	0.186	2.510	2.517	0.970	4.778	0.477
DiD-BCF (corrected)	50	3	0.179	2.449	2.455	0.970	4.810	0.505
DiD-BCF (corrected)	100	2	0.081	1.252	1.255	0.940	2.963	0.604
DiD-BCF (corrected)	100	3	0.058	1.145	1.147	0.960	2.985	0.671
DiD-BCF (corrected)	200	2	0.109	1.034	1.040	0.960	1.676	0.411
DiD-BCF (corrected)	200	3	0.094	1.025	1.029	0.970	1.659	0.414
DiD-BCF (corrected)	400	2	-0.200	1.303	1.319	0.960	1.482	0.297
DiD-BCF (corrected)	400	3	-0.194	1.242	1.257	0.960	1.455	0.305
DiD-BCF (corrected)	800	2	-0.046	0.315	0.318	0.990	0.624	0.509
DiD-BCF (corrected)	800	3	-0.049	0.335	0.339	0.980	0.643	0.484

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Table 6 continued from the previous page.

Estimator	N	d	Bias	SD(err)	RMSE	Cov.95	Len.95	$\overline{\text{sd}}/\text{SD}$
Callaway–Sant’Anna	50	2	0.042	0.460	0.462	0.948	2.040	0.958
Callaway–Sant’Anna	50	3	0.042	0.460	0.462	0.969	2.044	1.002
Callaway–Sant’Anna	100	2	0.027	0.309	0.311	0.960	1.307	1.056
Callaway–Sant’Anna	100	3	0.027	0.309	0.311	0.949	1.310	1.087
Callaway–Sant’Anna	200	2	-0.034	0.223	0.225	0.950	0.904	1.008
Callaway–Sant’Anna	200	3	-0.034	0.223	0.225	0.960	0.901	0.929
Callaway–Sant’Anna	400	2	0.007	0.150	0.150	0.950	0.635	0.995
Callaway–Sant’Anna	400	3	0.007	0.150	0.150	0.950	0.634	1.048
Callaway–Sant’Anna	800	2	0.001	0.108	0.108	0.980	0.449	0.902
Callaway–Sant’Anna	800	3	0.001	0.108	0.108	0.980	0.445	0.819
DoubleML DR-DiD	50	2	0.073	0.669	0.673	0.980	3.201	1.100
DoubleML DR-DiD	50	3	-0.024	0.737	0.737	0.990	3.556	1.172
DoubleML DR-DiD	100	2	0.077	0.441	0.448	1.000	1.976	1.137
DoubleML DR-DiD	100	3	-0.011	0.437	0.437	0.990	1.887	1.099
DoubleML DR-DiD	200	2	-0.045	0.265	0.269	0.960	1.172	1.102
DoubleML DR-DiD	200	3	-0.031	0.284	0.286	0.960	1.160	0.984
DoubleML DR-DiD	400	2	0.001	0.171	0.171	0.980	0.782	1.113
DoubleML DR-DiD	400	3	0.008	0.172	0.172	0.980	0.764	1.129
DoubleML DR-DiD	800	2	-0.000	0.113	0.113	0.990	0.515	0.966
DoubleML DR-DiD	800	3	0.003	0.124	0.124	0.950	0.516	0.860
DoubleML DR-DiD	1600	2	0.012	0.071	0.072	0.980	0.349	1.046
DoubleML DR-DiD	1600	3	0.008	0.072	0.073	1.000	0.353	1.021
grf-DiD (Wang)	50	2	0.007	0.412	0.412	0.920	1.427	0.751
grf-DiD (Wang)	50	3	0.023	0.409	0.410	0.900	1.443	0.860
grf-DiD (Wang)	100	2	0.009	0.304	0.304	0.850	0.970	0.792
grf-DiD (Wang)	100	3	0.007	0.302	0.302	0.860	0.997	0.847
grf-DiD (Wang)	200	2	-0.037	0.220	0.223	0.810	0.630	0.730
grf-DiD (Wang)	200	3	-0.037	0.220	0.223	0.810	0.630	0.729
grf-DiD (Wang)	400	2	0.005	0.148	0.148	0.890	0.444	0.734
grf-DiD (Wang)	400	3	0.005	0.148	0.148	0.890	0.444	0.733
grf-DiD (Wang)	800	2	0.001	0.111	0.111	0.810	0.319	0.644
grf-DiD (Wang)	800	3	0.001	0.111	0.111	0.810	0.319	0.645
grf-DiD (Wang)	1600	2	0.009	0.067	0.068	0.900	0.228	0.766
grf-DiD (Wang)	1600	3	0.009	0.067	0.068	0.900	0.228	0.766
<i>B2_sweep_serial</i>								
DiD-BCF (plain)	50	2	-0.028	0.278	0.280	0.910	0.935	0.676
DiD-BCF (plain)	50	3	-0.042	0.302	0.305	0.880	0.961	0.792
DiD-BCF (plain)	100	2	0.001	0.186	0.186	0.930	0.668	0.794
DiD-BCF (plain)	100	3	-0.010	0.188	0.188	0.940	0.679	0.809
DiD-BCF (plain)	200	2	-0.013	0.127	0.128	0.960	0.478	0.763
DiD-BCF (plain)	200	3	-0.025	0.120	0.122	0.960	0.481	0.754
DiD-BCF (plain)	400	2	0.016	0.098	0.099	0.910	0.344	0.745
DiD-BCF (plain)	400	3	0.011	0.096	0.097	0.920	0.348	0.772
DiD-BCF (plain)	800	2	0.000	0.071	0.071	0.910	0.249	0.689
DiD-BCF (plain)	800	3	-0.005	0.070	0.070	0.920	0.249	0.634
DiD-BCF (corrected)	50	2	0.211	1.851	1.863	1.000	3.596	0.496
DiD-BCF (corrected)	50	3	0.240	1.596	1.614	0.980	3.592	0.581
DiD-BCF (corrected)	100	2	0.085	0.882	0.886	0.990	2.283	0.668
DiD-BCF (corrected)	100	3	0.065	0.549	0.552	0.980	2.155	0.984
DiD-BCF (corrected)	200	2	0.019	0.268	0.269	0.990	1.160	1.065
DiD-BCF (corrected)	200	3	-0.011	0.155	0.156	0.990	1.163	1.626
DiD-BCF (corrected)	400	2	-0.113	0.831	0.839	0.990	1.018	0.315
DiD-BCF (corrected)	400	3	-0.094	0.653	0.660	1.000	0.922	0.362
DiD-BCF (corrected)	800	2	-0.014	0.175	0.176	1.000	0.431	0.605
DiD-BCF (corrected)	800	3	-0.018	0.145	0.146	0.990	0.442	0.694
Callaway–Sant’Anna	50	2	0.000	0.313	0.313	0.958	1.454	0.920
Callaway–Sant’Anna	50	3	0.000	0.313	0.313	0.969	1.458	0.998
Callaway–Sant’Anna	100	2	0.019	0.188	0.189	0.990	0.951	1.168

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Table 6 continued from the previous page.

Estimator	N	d	Bias	SD(err)	RMSE	Cov.95	Len.95	$\overline{\text{sd}}/\text{SD}$
Callaway–Sant’Anna	100	3	0.019	0.188	0.189	0.990	0.961	1.198
Callaway–Sant’Anna	200	2	-0.028	0.132	0.135	0.990	0.660	1.063
Callaway–Sant’Anna	200	3	-0.028	0.132	0.135	0.990	0.668	0.994
Callaway–Sant’Anna	400	2	0.007	0.091	0.091	1.000	0.464	1.060
Callaway–Sant’Anna	400	3	0.007	0.091	0.091	0.990	0.466	1.126
Callaway–Sant’Anna	800	2	0.004	0.071	0.071	0.980	0.329	0.881
Callaway–Sant’Anna	800	3	0.004	0.071	0.071	0.980	0.326	0.772
DoubleML DR-DiD	50	2	0.084	0.609	0.615	0.990	2.504	0.949
DoubleML DR-DiD	50	3	0.019	0.453	0.454	1.000	2.146	1.124
DoubleML DR-DiD	100	2	0.013	0.295	0.295	0.970	1.291	1.109
DoubleML DR-DiD	100	3	-0.008	0.265	0.265	1.000	1.313	1.208
DoubleML DR-DiD	200	2	-0.023	0.185	0.186	0.970	0.814	1.002
DoubleML DR-DiD	200	3	-0.027	0.179	0.181	0.990	0.829	0.996
DoubleML DR-DiD	400	2	0.001	0.113	0.113	0.980	0.543	0.989
DoubleML DR-DiD	400	3	0.012	0.115	0.115	0.990	0.537	1.119
DoubleML DR-DiD	800	2	-0.004	0.076	0.076	1.000	0.366	0.930
DoubleML DR-DiD	800	3	0.004	0.075	0.075	1.000	0.369	0.873
DoubleML DR-DiD	1600	2	0.008	0.048	0.049	1.000	0.251	0.940
DoubleML DR-DiD	1600	3	0.006	0.047	0.047	0.990	0.251	0.999
grf-DiD (Wang)	50	2	-0.013	0.291	0.291	0.940	1.093	0.757
grf-DiD (Wang)	50	3	0.004	0.284	0.284	0.960	1.119	0.929
grf-DiD (Wang)	100	2	0.012	0.188	0.188	0.900	0.730	0.899
grf-DiD (Wang)	100	3	0.006	0.188	0.188	0.900	0.766	0.979
grf-DiD (Wang)	200	2	-0.032	0.130	0.133	0.900	0.458	0.809
grf-DiD (Wang)	200	3	-0.032	0.129	0.133	0.910	0.458	0.811
grf-DiD (Wang)	400	2	0.004	0.090	0.090	0.910	0.334	0.829
grf-DiD (Wang)	400	3	0.004	0.090	0.090	0.910	0.334	0.830
grf-DiD (Wang)	800	2	0.007	0.073	0.074	0.880	0.244	0.672
grf-DiD (Wang)	800	3	0.006	0.074	0.074	0.890	0.244	0.671
grf-DiD (Wang)	1600	2	0.008	0.042	0.043	0.970	0.173	0.793
grf-DiD (Wang)	1600	3	0.008	0.043	0.043	0.970	0.173	0.791

Note. 100 replications per cell; the $d = 1$ counterpart is Table 5. TWFE and Synthetic DiD use no covariates, so their entries are invariant to d by construction. DoubleML DR-DiD is available at all 5 panel sizes in both sweeps.

7 D staggered adoption results

7.1 Cohort-time and event-time effects

The D results are the cleanest current test of the structured restriction. Table 7 reports event-study effects, and Table 8 reports the cohort-time comparisons. In the headline staggered design, the structured plain and corrected DiD-BCF estimates have small bias at the early event times and remain substantially closer to the true event-time path than pooled TWFE. The corrected method has near-nominal coverage across the displayed event times. The plain method is more efficient in point estimates but loses coverage at the latest event time, where the effect is most heterogeneous. Both tables in this subsection are $d = 1$ only; the nonlinear degrees are reported separately below and change the conclusion about the correction materially.

The contamination design magnifies the TWFE problem. TWFE understates the later event-time effects, and its bias grows with the stronger dynamic ramp. The structured DiD-BCF estimates continue to track the cohort-time targets, with the correction giving close-to-nominal coverage in the current cells. Gardner, Callaway–Sant’Anna, DoubleML, and grf-DiD are also strong on their respective cohort-time or event-time targets, so the relevant conclusion is not that DiD-BCF dominates every scalar benchmark. The conclusion is that pooling the D design into a single TWFE coefficient loses the very heterogeneity that the DGP was designed to reveal.

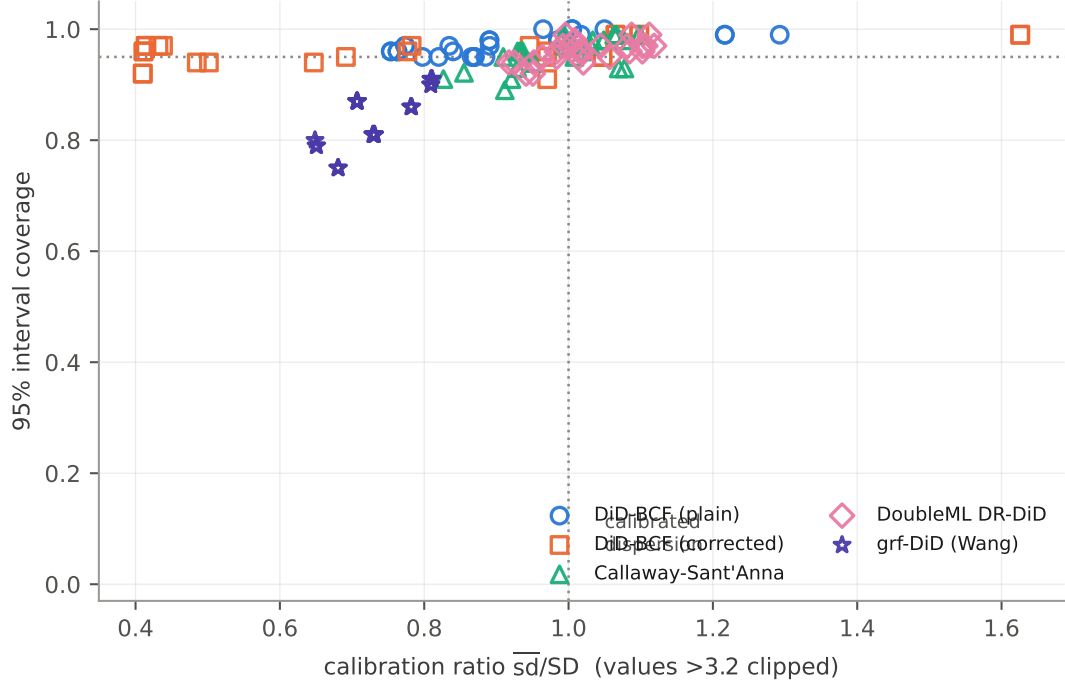


Figure 3: Posterior-spread calibration across the current B1, B2, and D settings and degrees. A ratio below one indicates an average reported spread smaller than the Monte-Carlo error dispersion.

Table 7: Event-study estimands under staggered adoption with cohort- and event-time-varying effects ($d = 1$, $N = 200$, 100 replications).

Estimator	k	True $ATT(k)$	Bias	SD(err)	RMSE	Cov.95
<i>D_staggered</i>						
DiD-BCF (plain)	0	4.072	0.094	0.114	0.148	0.850
DiD-BCF (plain)	1	5.701	-0.022	0.124	0.126	0.970
DiD-BCF (plain)	2	6.102	-0.044	0.158	0.164	0.960
DiD-BCF (plain)	3	6.046	-0.363	0.225	0.428	0.660
DiD-BCF (corrected)	0	4.072	-0.011	0.159	0.159	1.000
DiD-BCF (corrected)	1	5.701	-0.015	0.191	0.191	0.990
DiD-BCF (corrected)	2	6.102	-0.003	0.245	0.245	1.000
DiD-BCF (corrected)	3	6.046	-0.004	0.393	0.393	0.970
TWFE	0	4.069	0.196	0.146	0.245	0.930
TWFE	1	5.696	-0.284	0.186	0.340	0.840
TWFE	2	6.103	-0.460	0.255	0.526	0.830
TWFE	3	6.041	-0.656	0.380	0.758	0.810
Gardner (did2s)	0	4.072	-0.008	0.140	0.140	1.000
Gardner (did2s)	1	5.701	-0.020	0.177	0.178	1.000
Gardner (did2s)	2	6.102	0.005	0.241	0.241	1.000
Gardner (did2s)	3	6.046	-0.032	0.363	0.365	0.970
<i>D_contamination</i>						
DiD-BCF (plain)	0	5.389	0.135	0.118	0.179	0.810
DiD-BCF (plain)	1	9.700	0.004	0.130	0.130	0.980
DiD-BCF (plain)	2	10.482	0.055	0.172	0.180	0.940
DiD-BCF (plain)	3	9.344	-0.421	0.236	0.482	0.570
DiD-BCF (corrected)	0	5.389	-0.026	0.159	0.161	1.000
DiD-BCF (corrected)	1	9.700	-0.026	0.200	0.201	1.000
DiD-BCF (corrected)	2	10.482	-0.023	0.246	0.248	1.000

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Table 7 continued from the previous page.

Estimator	k	True $ATT(k)$	Bias	SD(err)	RMSE	Cov.95
DiD-BCF (corrected)	3	9.344	-0.002	0.404	0.404	1.000
TWFE	0	5.384	0.600	0.167	0.623	0.340
TWFE	1	9.691	-0.561	0.267	0.622	0.760
TWFE	2	10.485	-1.032	0.340	1.087	0.580
TWFE	3	9.336	-1.739	0.487	1.806	0.190
Gardner (did2s)	0	5.389	-0.008	0.140	0.140	1.000
Gardner (did2s)	1	9.700	-0.020	0.177	0.178	1.000
Gardner (did2s)	2	10.482	0.005	0.241	0.241	1.000
Gardner (did2s)	3	9.344	-0.032	0.363	0.365	0.990

Note. The true effect grows with exposure k and differs across cohorts, so $ATT(k)$ is not constant. Callaway–Sant’Anna, DoubleML and grf-DiD report cohort-by-time rather than event-time effects; they are summarised over cells in Table 8 and given cell by cell, ordered by event time, in Table 10. The nonlinear degrees are in Table 9.

Table 8: Cohort-by-time estimands $GATT(g, t)$ under staggered adoption, averaged over the estimated cells ($N = 200$, all three linearity degrees).

Estimator	d	Cells	True	Bias	SD(err)	RMSE	Cov.95
<i>D_staggered</i>							
DiD-BCF (plain)	1	9	5.355	-0.033	0.179	0.319	0.739
DiD-BCF (corrected)	1	9	5.355	-0.011	0.354	0.358	0.983
Callaway–Sant’Anna	1	9	5.355	-0.017	0.296	0.298	0.993
DoubleML DR-DiD	1	9	5.352	-0.008	0.410	0.412	0.979
grf-DiD (Wang)	1	9	5.352	-0.012	0.299	0.301	0.949
DiD-BCF (plain)	2	9	5.355	-0.041	0.190	0.329	0.729
DiD-BCF (corrected)	2	9	5.355	0.259	3.442	3.455	0.986
Callaway–Sant’Anna	2	9	5.356	-0.014	0.302	0.304	0.985
DoubleML DR-DiD	2	9	5.352	-0.027	0.407	0.409	0.982
grf-DiD (Wang)	2	9	5.352	-0.012	0.299	0.301	0.948
DiD-BCF (plain)	3	9	5.332	-0.075	0.196	0.340	0.723
DiD-BCF (corrected)	3	9	5.332	0.217	3.449	3.461	0.983
Callaway–Sant’Anna	3	9	5.330	-0.014	0.302	0.304	0.987
DoubleML DR-DiD	3	9	5.352	-0.030	0.430	0.433	0.983
grf-DiD (Wang)	3	9	5.352	-0.012	0.299	0.301	0.948
<i>D_contamination</i>							
DiD-BCF (plain)	1	9	8.570	0.006	0.218	0.441	0.604
DiD-BCF (corrected)	1	9	8.570	-0.024	0.354	0.358	0.994
Callaway–Sant’Anna	1	9	8.570	-0.017	0.296	0.298	0.994
DoubleML DR-DiD	1	9	8.565	-0.020	0.436	0.437	0.989
grf-DiD (Wang)	1	9	8.565	-0.010	0.311	0.313	0.963
DiD-BCF (plain)	2	9	8.570	-0.003	0.230	0.465	0.570
DiD-BCF (corrected)	2	9	8.570	0.122	2.587	2.595	0.990
Callaway–Sant’Anna	2	9	8.570	-0.014	0.302	0.304	0.992
DoubleML DR-DiD	2	9	8.565	-0.016	0.428	0.430	0.990
grf-DiD (Wang)	2	9	8.565	-0.010	0.311	0.313	0.964
DiD-BCF (plain)	3	9	8.533	-0.068	0.253	0.506	0.579
DiD-BCF (corrected)	3	9	8.533	0.048	2.846	2.851	0.990
Callaway–Sant’Anna	3	9	8.529	-0.014	0.302	0.304	0.990
DoubleML DR-DiD	3	9	8.565	-0.019	0.448	0.452	0.992
grf-DiD (Wang)	3	9	8.565	-0.010	0.311	0.313	0.963

Note. “Cells” is the number of (g, t) cells each estimator reports; all quantities are unweighted averages across those cells, which mixes cohorts and event times. The cell-by-cell version at $d = 1$ is Table 10.

7.2 Cell-level comparison and the nonlinear degrees

Table 8 averages over the nine estimated cells, which mixes cohorts and event times and can hide the cell in which an estimator fails. Table 10 therefore reports every (g, t) cell separately, grouped by event time. This is the only presentation in which Callaway–Sant’Anna, DoubleML, and grf-DiD are directly comparable with the DiD-BCF fits at a given event time, because collapsing their cells to a single event-time coefficient would require the cross-cohort covariance of the cell estimates, and the stored per-replication summaries do not carry it.

The cell table changes the reading of the plain fit. Its aggregate looks competitive, but the disaggregation shows that its error is concentrated in specific cells rather than spread evenly: in the headline design its bias is 0.223 at GATT(5, 5) and 0.170 at GATT(6, 6) with coverage of 0.76 and 0.85, while Callaway–Sant’Anna, DoubleML, and grf-DiD stay within roughly 0.05 of the truth in the same cells. The corrected fit removes most of that cell-specific error. The pattern is that the later-adopting cohorts, which have the shortest pre-period and the smallest effective control group, are where the plain posterior is both biased and over-confident, and this is exactly the diagnostic the averaged table suppresses.

Table 9 extends the event-study comparison to $d = 2$ and $d = 3$, and it overturns the reading that the $d = 1$ table alone would support. The plain fit is almost unaffected by the degree: its event-study RMSE moves from 0.148–0.428 at $d = 1$ to 0.146–0.498 at $d = 3$ in the headline design. The corrected fit collapses. Its RMSE rises from 0.159–0.393 at $d = 1$ to between 1.370 and 3.626 at $d = 2$ and $d = 3$ in the headline design, and to between 0.852 and 2.864 in the contamination design. At those degrees the corrected event-study estimates are worse than pooled TWFE at every event time, even though TWFE is the estimator this section is designed to reject. The conclusion that the posterior correction is the safer choice under staggered adoption is therefore specific to $d = 1$ and must not be stated for the design as a whole. This is the same failure mode already visible in the B1 and B2 nonlinear cells, and it is the strongest argument for treating the current correction implementation as provisional.

Figure 4 then places every estimator on a common bias scale against event time. Levels cannot mix the two estimator families, since the event-time target of the estimators that report $ES(k)$ directly is weighted by realised cohort sizes, whereas a cohort-time estimator can only be collapsed by averaging its cells, and the two targets differ by up to 0.24 in the contamination design. Bias is immune to that mismatch, because the mean of the per-cell biases at event time k is exactly the bias of the cell-averaged estimate, so each estimator is scored against its own target and all of them share the zero line. Read that way, the contamination panel is unambiguous: TWFE runs from +0.60 at $k = 0$ to -1.74 at $k = 3$, while DoubleML, grf-DiD, Callaway–Sant’Anna, Gardner, and corrected DiD-BCF all remain inside ± 0.06 at every event time, and plain DiD-BCF drifts to -0.42 at $k = 3$.

Table 9: Event-study estimands under staggered adoption at the nonlinear degrees $d = 2$ and $d = 3$ ($N = 200$, 100 replications).

Estimator	d	k	True ATT(k)	Bias	SD(err)	RMSE	Cov.95
<i>D.staggered</i>							
DiD-BCF (plain)	2	0	4.072	0.093	0.118	0.151	0.860
DiD-BCF (plain)	2	1	5.701	-0.031	0.130	0.134	0.980
DiD-BCF (plain)	2	2	6.102	-0.058	0.163	0.173	0.950
DiD-BCF (plain)	2	3	6.046	-0.381	0.233	0.446	0.650
DiD-BCF (plain)	3	0	4.050	0.080	0.122	0.146	0.900
DiD-BCF (plain)	3	1	5.670	-0.065	0.128	0.144	0.960
DiD-BCF (plain)	3	2	6.091	-0.109	0.171	0.203	0.920
DiD-BCF (plain)	3	3	6.020	-0.432	0.248	0.498	0.590
DiD-BCF (corrected)	2	0	4.072	0.191	2.050	2.059	1.000
DiD-BCF (corrected)	2	1	5.701	0.178	1.813	1.822	0.990

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Table 9 continued from the previous page.

Estimator	d	k	True ATT(k)	Bias	SD(err)	RMSE	Cov.95
DiD-BCF (corrected)	2	2	6.102	0.345	3.609	3.626	0.990
DiD-BCF (corrected)	2	3	6.046	0.267	3.351	3.362	0.980
DiD-BCF (corrected)	3	0	4.050	0.238	2.535	2.547	1.000
DiD-BCF (corrected)	3	1	5.670	0.103	1.366	1.370	1.000
DiD-BCF (corrected)	3	2	6.091	0.247	3.188	3.197	0.990
DiD-BCF (corrected)	3	3	6.020	0.155	2.590	2.595	0.990
TWFE	2	0	4.069	0.196	0.146	0.245	0.930
TWFE	2	1	5.696	-0.284	0.186	0.340	0.840
TWFE	2	2	6.103	-0.460	0.255	0.526	0.830
TWFE	2	3	6.041	-0.656	0.380	0.758	0.810
TWFE	3	0	4.069	0.196	0.146	0.245	0.930
TWFE	3	1	5.696	-0.284	0.186	0.340	0.840
TWFE	3	2	6.103	-0.460	0.255	0.526	0.830
TWFE	3	3	6.041	-0.656	0.380	0.758	0.810
Gardner (did2s)	2	0	4.072	-0.008	0.140	0.140	1.000
Gardner (did2s)	2	1	5.701	-0.020	0.177	0.178	1.000
Gardner (did2s)	2	2	6.102	0.005	0.241	0.241	1.000
Gardner (did2s)	2	3	6.046	-0.032	0.363	0.365	0.970
Gardner (did2s)	3	0	4.050	-0.008	0.140	0.140	1.000
Gardner (did2s)	3	1	5.670	-0.020	0.177	0.178	1.000
Gardner (did2s)	3	2	6.091	0.005	0.241	0.241	1.000
Gardner (did2s)	3	3	6.020	-0.032	0.363	0.365	0.960
<i>D_contamination</i>							
DiD-BCF (plain)	2	0	5.389	0.137	0.122	0.183	0.790
DiD-BCF (plain)	2	1	9.700	-0.004	0.138	0.138	0.960
DiD-BCF (plain)	2	2	10.482	0.040	0.176	0.180	0.930
DiD-BCF (plain)	2	3	9.344	-0.451	0.247	0.514	0.500
DiD-BCF (plain)	3	0	5.357	0.126	0.124	0.177	0.800
DiD-BCF (plain)	3	1	9.643	-0.075	0.162	0.179	0.940
DiD-BCF (plain)	3	2	10.467	-0.063	0.219	0.228	0.940
DiD-BCF (plain)	3	3	9.304	-0.559	0.316	0.642	0.440
DiD-BCF (corrected)	2	0	5.389	0.023	1.229	1.229	0.990
DiD-BCF (corrected)	2	1	9.700	0.081	1.229	1.231	1.000
DiD-BCF (corrected)	2	2	10.482	0.196	2.857	2.864	1.000
DiD-BCF (corrected)	2	3	9.344	0.202	2.075	2.085	1.000
DiD-BCF (corrected)	3	0	5.357	0.058	1.586	1.587	0.990
DiD-BCF (corrected)	3	1	9.643	-0.024	0.852	0.852	1.000
DiD-BCF (corrected)	3	2	10.467	0.018	2.338	2.339	1.000
DiD-BCF (corrected)	3	3	9.304	0.037	1.838	1.838	1.000
TWFE	2	0	5.384	0.600	0.167	0.623	0.340
TWFE	2	1	9.691	-0.561	0.267	0.622	0.760
TWFE	2	2	10.485	-1.032	0.340	1.087	0.580
TWFE	2	3	9.336	-1.739	0.487	1.806	0.190
TWFE	3	0	5.384	0.600	0.167	0.623	0.340
TWFE	3	1	9.691	-0.561	0.267	0.622	0.760
TWFE	3	2	10.485	-1.032	0.340	1.087	0.580
TWFE	3	3	9.336	-1.739	0.487	1.806	0.190
Gardner (did2s)	2	0	5.389	-0.008	0.140	0.140	1.000
Gardner (did2s)	2	1	9.700	-0.020	0.177	0.178	1.000
Gardner (did2s)	2	2	10.482	0.005	0.241	0.241	1.000
Gardner (did2s)	2	3	9.344	-0.032	0.363	0.365	0.990
Gardner (did2s)	3	0	5.357	-0.008	0.140	0.140	1.000
Gardner (did2s)	3	1	9.643	-0.020	0.177	0.178	1.000
Gardner (did2s)	3	2	10.467	0.005	0.241	0.241	1.000
Gardner (did2s)	3	3	9.304	-0.032	0.363	0.365	1.000

Note. The $d = 1$ counterpart is Table 7. TWFE uses no covariates, so its entries are invariant to d by construction; at $d = 3$ the treatment-effect surface itself becomes quadratic, so the true ATT(k) path changes as well.

Table 10: Cell-by-cell cohort-time estimands $GATT(g, t)$ under staggered adoption, grouped by event time $k = t - g$ ($d = 1$, $N = 200$, 100 replications).

Estimator	g	t	k	True	Bias	RMSE	Cov.95
<i>D_staggered</i>							
$k = 0$, cohort $g = 4$							
DiD-BCF (plain)	4	4	0	2.748	-0.087	0.180	0.970
DiD-BCF (corrected)	4	4	0	2.748	-0.010	0.357	0.950
Callaway-Sant'Anna	4	4	0	2.748	-0.049	0.303	0.960
DoubleML DR-DiD	4	4	0	2.746	-0.065	0.416	0.940
grf-DiD (Wang)	4	4	0	2.746	-0.050	0.300	0.870
$k = 0$, cohort $g = 5$							
DiD-BCF (plain)	5	5	0	4.124	0.223	0.274	0.760
DiD-BCF (corrected)	5	5	0	4.124	-0.086	0.365	0.960
Callaway-Sant'Anna	5	5	0	4.124	-0.031	0.292	0.990
DoubleML DR-DiD	5	5	0	4.126	0.011	0.417	0.950
grf-DiD (Wang)	5	5	0	4.126	-0.020	0.284	0.940
$k = 0$, cohort $g = 6$							
DiD-BCF (plain)	6	6	0	5.537	0.170	0.237	0.850
DiD-BCF (corrected)	6	6	0	5.537	0.059	0.298	1.000
Callaway-Sant'Anna	6	6	0	5.537	0.038	0.274	1.000
DoubleML DR-DiD	6	6	0	5.527	0.006	0.394	0.980
grf-DiD (Wang)	6	6	0	5.527	0.043	0.267	1.000
$k = 1$, cohort $g = 4$							
DiD-BCF (plain)	4	5	1	3.847	0.230	0.275	0.790
DiD-BCF (corrected)	4	5	1	3.847	-0.064	0.347	0.990
Callaway-Sant'Anna	4	5	1	3.847	-0.048	0.290	0.990
DoubleML DR-DiD	4	5	1	3.844	-0.043	0.383	0.980
grf-DiD (Wang)	4	5	1	3.844	-0.053	0.294	0.900
$k = 1$, cohort $g = 5$							
DiD-BCF (plain)	5	6	1	5.773	0.138	0.215	0.900
DiD-BCF (corrected)	5	6	1	5.773	-0.055	0.353	0.990
Callaway-Sant'Anna	5	6	1	5.773	-0.003	0.303	1.000
DoubleML DR-DiD	5	6	1	5.777	0.018	0.401	1.000
grf-DiD (Wang)	5	6	1	5.777	0.002	0.317	0.960
$k = 1$, cohort $g = 6$							
DiD-BCF (plain)	6	7	1	7.752	-0.475	0.519	0.360
DiD-BCF (corrected)	6	7	1	7.752	0.083	0.383	1.000
Callaway-Sant'Anna	6	7	1	7.752	0.005	0.316	1.000
DoubleML DR-DiD	6	7	1	7.738	0.007	0.443	1.000
grf-DiD (Wang)	6	7	1	7.738	0.022	0.315	1.000
$k = 2$, cohort $g = 4$							
DiD-BCF (plain)	4	6	2	4.947	0.251	0.301	0.750
DiD-BCF (corrected)	4	6	2	4.947	-0.043	0.369	0.990
Callaway-Sant'Anna	4	6	2	4.947	-0.026	0.309	1.000
DoubleML DR-DiD	4	6	2	4.943	-0.015	0.408	0.970
grf-DiD (Wang)	4	6	2	4.943	-0.020	0.331	0.920
$k = 2$, cohort $g = 5$							
DiD-BCF (plain)	5	7	2	7.423	-0.384	0.440	0.610
DiD-BCF (corrected)	5	7	2	7.423	0.021	0.360	1.000
Callaway-Sant'Anna	5	7	2	7.423	0.009	0.295	1.000
DoubleML DR-DiD	5	7	2	7.427	0.061	0.441	1.000
grf-DiD (Wang)	5	7	2	7.427	0.020	0.274	0.990
$k = 3$, cohort $g = 4$							
DiD-BCF (plain)	4	7	3	6.046	-0.363	0.428	0.660
DiD-BCF (corrected)	4	7	3	6.046	-0.004	0.393	0.970
Callaway-Sant'Anna	4	7	3	6.046	-0.044	0.298	1.000
DoubleML DR-DiD	4	7	3	6.041	-0.050	0.402	0.990
grf-DiD (Wang)	4	7	3	6.041	-0.050	0.331	0.960

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Table 10 continued from the previous page.

Estimator	g	t	k	True	Bias	RMSE	Cov.95
<i>D_contamination</i>							
$k = 0$, cohort $g = 4$							
DiD-BCF (plain)	4	4	0	2.748	-0.085	0.189	0.970
DiD-BCF (corrected)	4	4	0	2.748	-0.009	0.356	0.960
Callaway-Sant'Anna	4	4	0	2.748	-0.049	0.303	0.960
DoubleML DR-DiD	4	4	0	2.746	-0.047	0.404	0.970
grf-DiD (Wang)	4	4	0	2.746	-0.050	0.300	0.870
$k = 0$, cohort $g = 5$							
DiD-BCF (plain)	5	5	0	5.498	0.029	0.200	0.950
DiD-BCF (corrected)	5	5	0	5.498	-0.061	0.350	1.000
Callaway-Sant'Anna	5	5	0	5.498	-0.031	0.292	1.000
DoubleML DR-DiD	5	5	0	5.502	-0.013	0.393	0.990
grf-DiD (Wang)	5	5	0	5.502	-0.021	0.289	0.950
$k = 0$, cohort $g = 6$							
DiD-BCF (plain)	6	6	0	8.306	0.494	0.538	0.420
DiD-BCF (corrected)	6	6	0	8.306	-0.014	0.281	1.000
Callaway-Sant'Anna	6	6	0	8.306	0.038	0.274	1.000
DoubleML DR-DiD	6	6	0	8.291	0.015	0.353	1.000
grf-DiD (Wang)	6	6	0	8.291	0.050	0.277	1.000
$k = 1$, cohort $g = 4$							
DiD-BCF (plain)	4	5	1	4.947	0.033	0.186	0.980
DiD-BCF (corrected)	4	5	1	4.947	-0.045	0.347	0.990
Callaway-Sant'Anna	4	5	1	4.947	-0.048	0.290	0.990
DoubleML DR-DiD	4	5	1	4.943	-0.039	0.463	0.970
grf-DiD (Wang)	4	5	1	4.943	-0.056	0.296	0.930
$k = 1$, cohort $g = 5$							
DiD-BCF (plain)	5	6	1	9.897	0.554	0.602	0.360
DiD-BCF (corrected)	5	6	1	9.897	-0.090	0.354	1.000
Callaway-Sant'Anna	5	6	1	9.897	-0.003	0.303	1.000
DoubleML DR-DiD	5	6	1	9.903	-0.013	0.478	1.000
grf-DiD (Wang)	5	6	1	9.903	0.007	0.345	0.980
$k = 1$, cohort $g = 6$							
DiD-BCF (plain)	6	7	1	14.951	-0.580	0.637	0.340
DiD-BCF (corrected)	6	7	1	14.951	0.062	0.396	1.000
Callaway-Sant'Anna	6	7	1	14.951	0.005	0.316	1.000
DoubleML DR-DiD	6	7	1	14.923	-0.017	0.441	1.000
grf-DiD (Wang)	6	7	1	14.923	0.024	0.326	1.000
$k = 2$, cohort $g = 4$							
DiD-BCF (plain)	4	6	2	7.145	0.533	0.576	0.370
DiD-BCF (corrected)	4	6	2	7.145	-0.086	0.370	1.000
Callaway-Sant'Anna	4	6	2	7.145	-0.026	0.309	1.000
DoubleML DR-DiD	4	6	2	7.139	-0.046	0.448	0.990
grf-DiD (Wang)	4	6	2	7.139	-0.019	0.334	0.950
$k = 2$, cohort $g = 5$							
DiD-BCF (plain)	5	7	2	14.295	-0.500	0.560	0.480
DiD-BCF (corrected)	5	7	2	14.295	0.030	0.361	1.000
Callaway-Sant'Anna	5	7	2	14.295	0.009	0.295	1.000
DoubleML DR-DiD	5	7	2	14.304	0.038	0.471	1.000
grf-DiD (Wang)	5	7	2	14.304	0.021	0.309	1.000
$k = 3$, cohort $g = 4$							
DiD-BCF (plain)	4	7	3	9.344	-0.421	0.482	0.570
DiD-BCF (corrected)	4	7	3	9.344	-0.002	0.404	1.000
Callaway-Sant'Anna	4	7	3	9.344	-0.044	0.298	1.000
DoubleML DR-DiD	4	7	3	9.336	-0.055	0.483	0.980
grf-DiD (Wang)	4	7	3	9.336	-0.047	0.343	0.990

Note. Every estimator here reports the same (g, t) cell, so the comparison is exact and requires no aggregation. TWFE and Gardner are absent because they report only pooled or event-time coefficients; see Table 7. Reading

down a fixed k block shows how each estimator degrades as the cohort adopts later and its control group shrinks.

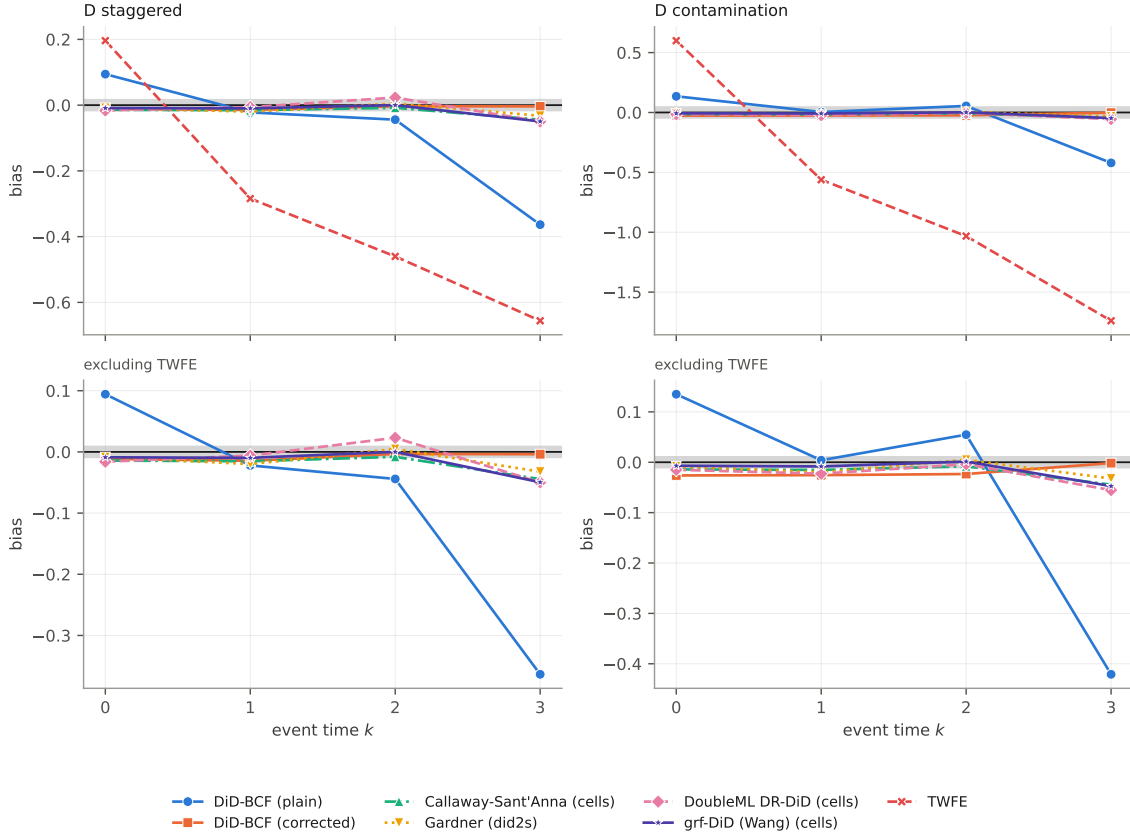


Figure 4: Bias against event time for every estimator in the two staggered designs at $d = 1$. The lower row repeats the upper row without TWFE so that the remaining estimators separate. Estimators marked “cells” report cohort-time effects and are summarised by the mean of their per-cell biases at each event time.

7.3 Goodman–Bacon decomposition

The decomposition in Table 11 makes the contamination mechanism explicit. The comparisons that use an earlier-treated cohort as a control for a later-treated cohort receive nonzero weight. Their combined weight is the same design feature in both D variants, but the dynamic and cohort multipliers make their contribution increasingly inconsistent with the target as the contamination ramp increases. The TWFE bias therefore moves from approximately -0.47 in the headline design to approximately -0.98 in the contamination variant.

8 Individual CATT-surface results

The paper’s distinctive target is an individual heterogeneous effect surface, not only a scalar ATT. Table 12 reports surface RMSE, MAE, MAPE, and pointwise interval coverage for every available scenario and degree. A GATT-only estimator is broadcast to treated observations in the relevant cell, which gives it the natural constant-prediction benchmark.

The surface results show three patterns. First, the plain structured DiD-BCF surface is substantially better than the broadcast scalar benchmarks in the canonical designs. Second, grf-DiD is the strongest non-DiD-BCF individual surface benchmark in most cells. Third, $d = 3$ increases the difficulty of the surface even when scalar bias remains small, because the treatment

Table 11: Goodman-Bacon (2021) decomposition of the TWFE estimate under staggered adoption with cohort- and event-time-varying effects, averaged over 500 replications.

Comparison type	Weight	$\widehat{DD}$	Contribution
<i>D_staggered</i>			
Earlier vs Later	0.072	3.386	0.245
Later vs Earlier	0.039	4.038	0.156
Treated vs Untreated	0.222	5.581	1.241
<i>TWFE estimate</i>			4.830
<i>true ATT</i>			5.299
<i>TWFE bias</i>			-0.469
<i>weight on already-treated comparisons</i>			0.116
<i>D_contamination</i>			
Earlier vs Later	0.072	4.026	0.291
Later vs Earlier	0.039	5.873	0.227
Treated vs Untreated	0.222	9.154	2.035
<i>TWFE estimate</i>			7.448
<i>true ATT</i>			8.432
<i>TWFE bias</i>			-0.984
<i>weight on already-treated comparisons</i>			0.116

Note. “Earlier vs. later” and “later vs. earlier” are the two-by-two comparisons that use already-treated units as controls; their combined weight is reported in the last line of each block. Weights are normalised over all two-by-two comparisons including those against the never-treated group.

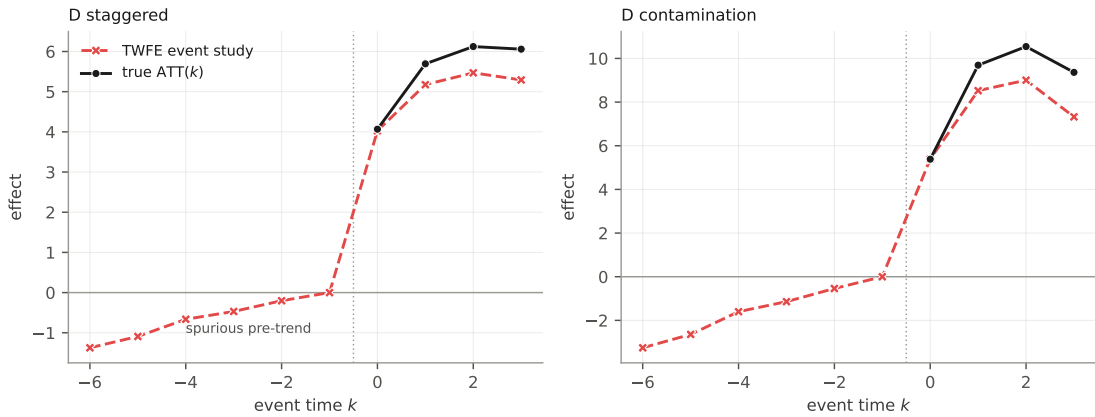


Figure 5: Event-study estimates and true event-time effects in the D designs.

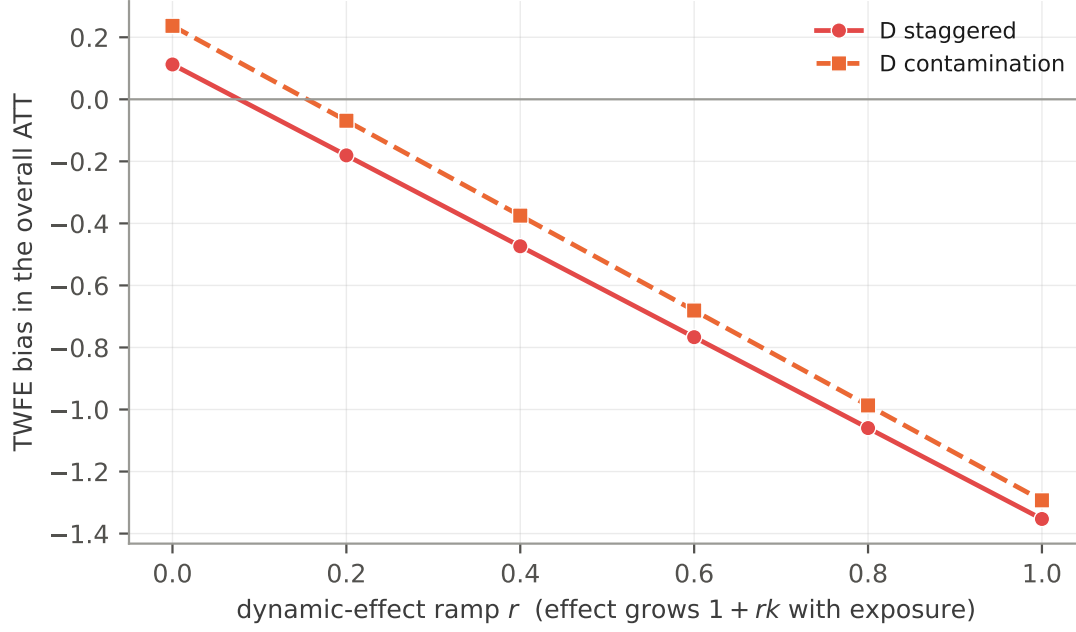


Figure 6: Effect paths under the headline staggered and contamination ramps.

effect is now quadratic and non-additive in the effect modifiers. The corrected surface is generally less efficient than the plain surface because the cell-level correction is broadcast to individual observations.

Table 12: CATT-surface accuracy: error of the estimated individual treatment effect over the treated observations, averaged across replications (Monte-Carlo SD of the within-replication RMSE in parentheses).

Estimator	d	RMSE (SD)	MAE	MAPE	Cov.95	Len.95
<i>B1_baseline</i>						
DiD-BCF (plain)	1	0.243 (0.056)	0.187	0.058	0.967	1.005
DiD-BCF (corrected)	1	1.071 (0.066)	0.880	0.285	0.316	0.977
TWFE	1	0.902 (0.041)	0.762	0.221	0.352	0.974
Callaway–Sant’Anna	1	1.069 (0.065)	0.878	0.285	0.292	0.898
Gardner (did2s)	1	1.071 (0.065)	0.882	0.284	0.320	0.996
DoubleML DR-DiD	1	0.914 (0.049)	0.770	0.225	0.404	1.153
Synthetic DiD	1	1.062 (0.063)	0.874	0.282	0.259	0.797
grf-DiD (Wang)	1	0.677 (0.093)	0.572	0.166	0.600	1.325
DiD-BCF (plain)	2	0.263 (0.065)	0.206	0.064	0.957	1.033
DiD-BCF (corrected)	2	1.234 (1.115)	1.034	0.330	0.347	1.851
TWFE	2	0.902 (0.041)	0.762	0.221	0.352	0.974
Callaway–Sant’Anna	2	1.071 (0.066)	0.880	0.285	0.302	0.934
Gardner (did2s)	2	1.071 (0.065)	0.882	0.284	0.320	0.996
DoubleML DR-DiD	2	0.916 (0.052)	0.772	0.226	0.399	1.138
Synthetic DiD	2	1.062 (0.063)	0.874	0.282	0.259	0.797
grf-DiD (Wang)	2	0.677 (0.093)	0.572	0.166	0.600	1.325
DiD-BCF (plain)	3	0.463 (0.079)	0.322	0.085	0.851	1.084
DiD-BCF (corrected)	3	1.238 (1.103)	1.016	0.294	0.342	1.837
TWFE	3	0.902 (0.041)	0.762	0.221	0.352	0.974
Callaway–Sant’Anna	3	1.082 (0.104)	0.869	0.250	0.298	0.931
Gardner (did2s)	3	1.082 (0.102)	0.871	0.250	0.312	0.997
DoubleML DR-DiD	3	0.915 (0.051)	0.771	0.225	0.399	1.136
Synthetic DiD	3	1.073 (0.101)	0.865	0.248	0.250	0.797
grf-DiD (Wang)	3	0.677 (0.093)	0.572	0.166	0.600	1.325

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Table 12 continued from the previous page.

Estimator	d	RMSE (SD)	MAE	MAPE	Cov.95	Len.95
<i>B1_strong_confounder</i>						
DiD-BCF (plain)	1	0.241 (0.056)	0.188	0.058	0.973	1.046
DiD-BCF (corrected)	1	1.068 (0.064)	0.877	0.285	0.320	0.986
TWFE	1	0.900 (0.044)	0.760	0.220	0.352	0.974
Callaway-Sant'Anna	1	1.064 (0.063)	0.875	0.284	0.294	0.899
Gardner (did2s)	1	1.070 (0.066)	0.880	0.283	0.320	0.995
DoubleML DR-DiD	1	0.915 (0.048)	0.770	0.224	0.406	1.149
Synthetic DiD	1	1.060 (0.063)	0.873	0.281	0.257	0.795
grf-DiD (Wang)	1	0.671 (0.090)	0.567	0.165	0.608	1.324
DiD-BCF (plain)	2	0.260 (0.064)	0.206	0.064	0.964	1.070
DiD-BCF (corrected)	2	1.111 (0.224)	0.915	0.296	0.362	1.348
TWFE	2	0.900 (0.044)	0.760	0.220	0.352	0.974
Callaway-Sant'Anna	2	1.066 (0.065)	0.876	0.284	0.304	0.932
Gardner (did2s)	2	1.070 (0.066)	0.880	0.283	0.320	0.995
DoubleML DR-DiD	2	0.913 (0.051)	0.769	0.225	0.404	1.146
Synthetic DiD	2	1.060 (0.063)	0.873	0.281	0.257	0.795
grf-DiD (Wang)	2	0.671 (0.090)	0.567	0.165	0.607	1.324
DiD-BCF (plain)	3	0.430 (0.076)	0.304	0.081	0.888	1.150
DiD-BCF (corrected)	3	1.119 (0.240)	0.902	0.261	0.362	1.352
TWFE	3	0.900 (0.044)	0.760	0.220	0.352	0.974
Callaway-Sant'Anna	3	1.077 (0.102)	0.863	0.249	0.303	0.931
Gardner (did2s)	3	1.080 (0.101)	0.868	0.249	0.315	0.995
DoubleML DR-DiD	3	0.912 (0.048)	0.768	0.224	0.406	1.148
Synthetic DiD	3	1.071 (0.100)	0.862	0.247	0.255	0.796
grf-DiD (Wang)	3	0.671 (0.090)	0.567	0.164	0.607	1.324
<i>B1_serial_corr</i>						
DiD-BCF (plain)	1	0.295 (0.075)	0.235	0.072	0.890	0.935
DiD-BCF (corrected)	1	1.063 (0.063)	0.875	0.284	0.263	0.806
TWFE	1	0.902 (0.044)	0.762	0.221	0.348	0.967
Callaway-Sant'Anna	1	1.063 (0.062)	0.875	0.284	0.250	0.771
Gardner (did2s)	1	1.071 (0.066)	0.882	0.284	0.318	0.989
DoubleML DR-DiD	1	0.903 (0.041)	0.763	0.223	0.348	0.966
Synthetic DiD	1	1.062 (0.061)	0.875	0.282	0.254	0.792
grf-DiD (Wang)	1	0.615 (0.098)	0.518	0.150	0.589	1.180
DiD-BCF (plain)	2	0.312 (0.087)	0.250	0.077	0.879	0.969
DiD-BCF (corrected)	2	1.130 (0.463)	0.930	0.301	0.295	1.401
TWFE	2	0.902 (0.044)	0.762	0.221	0.348	0.967
Callaway-Sant'Anna	2	1.065 (0.064)	0.877	0.284	0.261	0.809
Gardner (did2s)	2	1.071 (0.066)	0.882	0.284	0.318	0.989
DoubleML DR-DiD	2	0.903 (0.044)	0.764	0.224	0.345	0.962
Synthetic DiD	2	1.062 (0.061)	0.875	0.282	0.254	0.792
grf-DiD (Wang)	2	0.614 (0.097)	0.518	0.150	0.590	1.180
DiD-BCF (plain)	3	0.456 (0.095)	0.331	0.089	0.819	1.058
DiD-BCF (corrected)	3	1.158 (0.639)	0.929	0.270	0.290	1.455
TWFE	3	0.902 (0.044)	0.762	0.221	0.348	0.967
Callaway-Sant'Anna	3	1.076 (0.103)	0.865	0.249	0.256	0.806
Gardner (did2s)	3	1.082 (0.103)	0.869	0.250	0.311	0.990
DoubleML DR-DiD	3	0.901 (0.041)	0.762	0.222	0.346	0.960
Synthetic DiD	3	1.072 (0.101)	0.864	0.248	0.251	0.793
grf-DiD (Wang)	3	0.614 (0.098)	0.518	0.150	0.590	1.181
<i>B1_selection_obs</i>						
DiD-BCF (plain)	1	0.261 (0.058)	0.202	0.061	0.960	1.028
DiD-BCF (corrected)	1	1.070 (0.090)	0.870	0.270	0.408	1.256
TWFE	1	0.889 (0.061)	0.743	0.211	0.366	0.971
Callaway-Sant'Anna	1	1.062 (0.077)	0.864	0.268	0.370	1.083
Gardner (did2s)	1	1.204 (0.115)	0.959	0.327	0.326	0.972
DoubleML DR-DiD	1	0.912 (0.072)	0.758	0.215	0.464	1.287
Synthetic DiD	1	1.120 (0.091)	0.893	0.302	0.283	0.778

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Table 12 continued from the previous page.

Estimator	d	RMSE (SD)	MAE	MAPE	Cov.95	Len.95
grf-DiD (Wang)	1	0.691 (0.089)	0.583	0.161	0.607	1.370
DiD-BCF (plain)	2	0.286 (0.066)	0.225	0.067	0.946	1.052
DiD-BCF (corrected)	2	1.373 (1.456)	1.135	0.342	0.505	3.001
TWFE	2	0.889 (0.061)	0.743	0.211	0.366	0.971
Callaway–Sant’Anna	2	1.091 (0.129)	0.887	0.279	0.399	1.225
Gardner (did2s)	2	1.204 (0.115)	0.959	0.327	0.326	0.972
DoubleML DR-DiD	2	0.918 (0.072)	0.763	0.217	0.469	1.311
Synthetic DiD	2	1.120 (0.091)	0.893	0.302	0.283	0.778
grf-DiD (Wang)	2	0.691 (0.089)	0.583	0.161	0.607	1.370
DiD-BCF (plain)	3	0.485 (0.078)	0.350	0.092	0.824	1.100
DiD-BCF (corrected)	3	1.387 (1.492)	1.102	0.308	0.554	2.994
TWFE	3	0.889 (0.061)	0.743	0.211	0.366	0.971
Callaway–Sant’Anna	3	1.106 (0.144)	0.847	0.243	0.456	1.229
Gardner (did2s)	3	1.218 (0.134)	0.953	0.295	0.366	0.974
DoubleML DR-DiD	3	0.914 (0.074)	0.761	0.216	0.466	1.305
Synthetic DiD	3	1.135 (0.116)	0.863	0.265	0.370	0.780
grf-DiD (Wang)	3	0.691 (0.089)	0.583	0.161	0.607	1.370
<i>D_staggered</i>						
DiD-BCF (plain)	1	0.674 (0.068)	0.496	0.195	0.774	1.412
DiD-BCF (corrected)	1	2.123 (0.118)	1.682	0.786	0.310	1.745
TWFE	1	2.212 (0.091)	1.784	0.414	0.191	1.145
Callaway–Sant’Anna	1	2.113 (0.118)	1.674	0.786	0.285	1.620
Gardner (did2s)	1	2.466 (0.116)	1.968	0.969	0.192	1.245
DoubleML DR-DiD	1	1.799 (0.076)	1.464	0.332	0.378	1.935
Synthetic DiD	1	2.914 (0.138)	2.223	0.814	0.145	0.922
grf-DiD (Wang)	1	1.520 (0.068)	1.242	0.283	0.384	1.714
DiD-BCF (plain)	2	0.701 (0.073)	0.515	0.199	0.765	1.438
DiD-BCF (corrected)	2	2.692 (3.408)	2.066	0.871	0.351	3.846
TWFE	2	2.212 (0.091)	1.784	0.414	0.191	1.145
Callaway–Sant’Anna	2	2.113 (0.119)	1.674	0.789	0.290	1.651
Gardner (did2s)	2	2.466 (0.116)	1.968	0.969	0.192	1.245
DoubleML DR-DiD	2	1.798 (0.071)	1.464	0.331	0.378	1.947
Synthetic DiD	2	2.914 (0.138)	2.223	0.814	0.145	0.922
grf-DiD (Wang)	2	1.520 (0.069)	1.242	0.283	0.384	1.714
DiD-BCF (plain)	3	0.877 (0.134)	0.576	0.116	0.762	1.527
DiD-BCF (corrected)	3	2.690 (3.431)	2.031	0.457	0.352	3.821
TWFE	3	2.212 (0.091)	1.784	0.414	0.191	1.145
Callaway–Sant’Anna	3	2.120 (0.188)	1.636	0.368	0.293	1.653
Gardner (did2s)	3	2.464 (0.188)	1.943	0.459	0.161	1.243
DoubleML DR-DiD	3	1.804 (0.082)	1.468	0.331	0.381	1.965
Synthetic DiD	3	2.911 (0.200)	2.120	0.401	0.158	0.924
grf-DiD (Wang)	3	1.520 (0.069)	1.242	0.283	0.384	1.713
<i>D_contamination</i>						
DiD-BCF (plain)	1	0.909 (0.093)	0.659	0.169	0.788	1.932
DiD-BCF (corrected)	1	3.525 (0.205)	2.652	0.784	0.264	2.242
TWFE	1	4.435 (0.167)	3.521	0.580	0.130	1.674
Callaway–Sant’Anna	1	3.519 (0.206)	2.646	0.785	0.252	2.203
Gardner (did2s)	1	4.748 (0.201)	3.760	1.215	0.134	1.904
DoubleML DR-DiD	1	2.970 (0.122)	2.300	0.329	0.322	2.358
Synthetic DiD	1	5.600 (0.219)	4.131	1.074	0.098	1.133
grf-DiD (Wang)	1	2.402 (0.105)	1.851	0.269	0.357	2.218
DiD-BCF (plain)	2	0.953 (0.103)	0.690	0.166	0.777	1.969
DiD-BCF (corrected)	2	3.929 (2.143)	2.942	0.834	0.302	4.197
TWFE	2	4.435 (0.167)	3.521	0.580	0.130	1.674
Callaway–Sant’Anna	2	3.519 (0.213)	2.647	0.787	0.254	2.221
Gardner (did2s)	2	4.748 (0.201)	3.760	1.215	0.134	1.904
DoubleML DR-DiD	2	2.969 (0.125)	2.298	0.329	0.319	2.346
Synthetic DiD	2	5.600 (0.219)	4.131	1.074	0.098	1.133
grf-DiD (Wang)	2	2.402 (0.105)	1.851	0.269	0.356	2.217

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Table 12 continued from the previous page.

Estimator	d	RMSE (SD)	MAE	MAPE	Cov.95	Len.95
DiD-BCF (plain)	3	1.321 (0.229)	0.818	0.110	0.736	2.057
DiD-BCF (corrected)	3	3.969 (2.452)	2.908	0.416	0.295	4.239
TWFE	3	4.435 (0.167)	3.521	0.580	0.130	1.674
Callaway-Sant'Anna	3	3.527 (0.319)	2.588	0.367	0.256	2.227
Gardner (did2s)	3	4.737 (0.316)	3.675	0.612	0.122	1.898
DoubleML DR-DiD	3	2.972 (0.121)	2.299	0.329	0.323	2.376
Synthetic DiD	3	5.588 (0.323)	3.991	0.569	0.091	1.136
grf-DiD (Wang)	3	2.402 (0.105)	1.851	0.269	0.357	2.218

Note. $N = 200$, 100 replications. Estimators that produce only cell-level effects have their $GATT(g, t)$ (or event-study, or overall ATT) estimate broadcast to every treated observation in the cell, which is the natural best case for a homogeneous-effect estimator. Cov.95 is *pointwise* coverage of the individual effect.

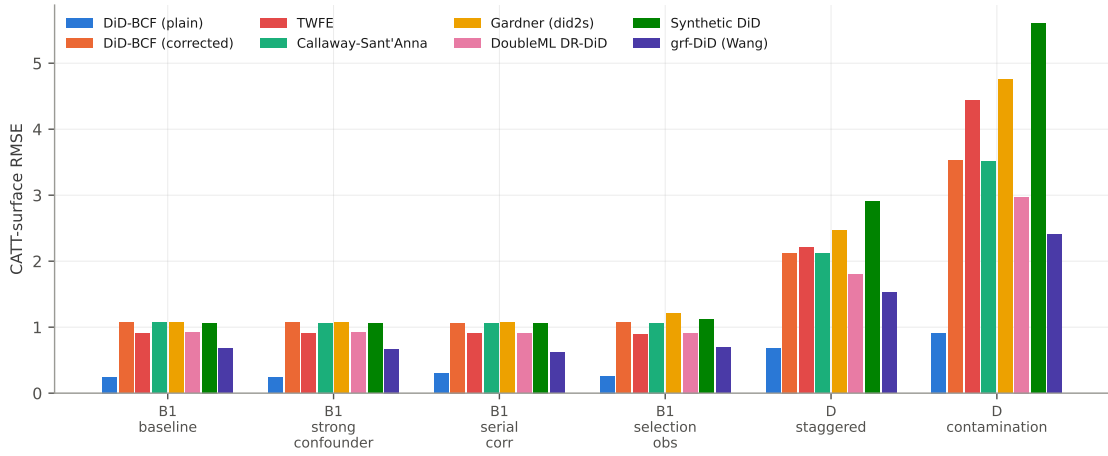


Figure 7: CATT-surface accuracy across the complete available grid.

9 Post-correction route audit: structured DiD-BCF and unit random effect

9.1 What is newly available

The route file contains the first direct comparison between the two structured implementations under a common sampling budget. It is a partial B2 ordinary sweep at $d = 1$ with 15 replications per cell. The benchmark rows below use the current 100-replication benchmark files at $N = 200$. These route results are not substitutes for the complete 100-replication grid, but they answer the implementation question the full grid cannot answer because it has no unit-random-intercept output.

The two route implementations are close in scalar point estimation. The unit-intercept version has the lower surface RMSE, which is consistent with the canonical DGP containing a persistent unobserved unit effect. The benchmark comparison confirms that the surface advantage is not simply a scalar ATT advantage: the GATT-only competitors are broadcast and the individual-level grf-DiD benchmark is better than those but remains above the two DiD-BCF route fits in this small partial run. Coverage is not decisive because 15 replications make a coverage estimate of 1.000 compatible with a Monte-Carlo standard error of about 0.055.

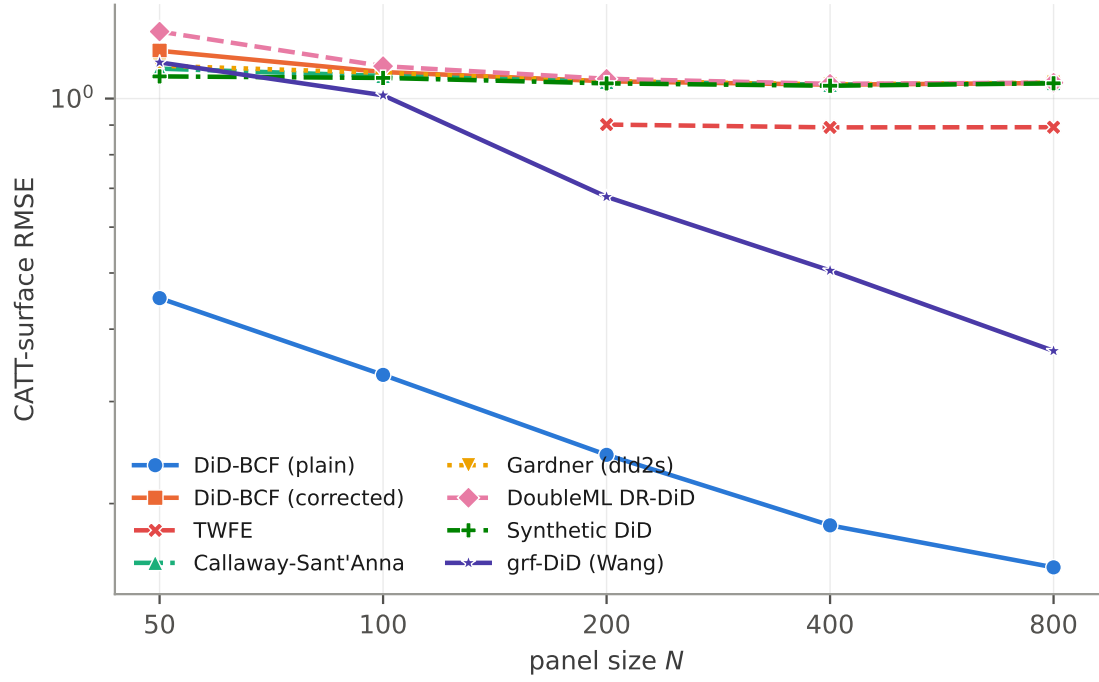


Figure 8: CATT-surface accuracy over the available B2 panel-size cells.

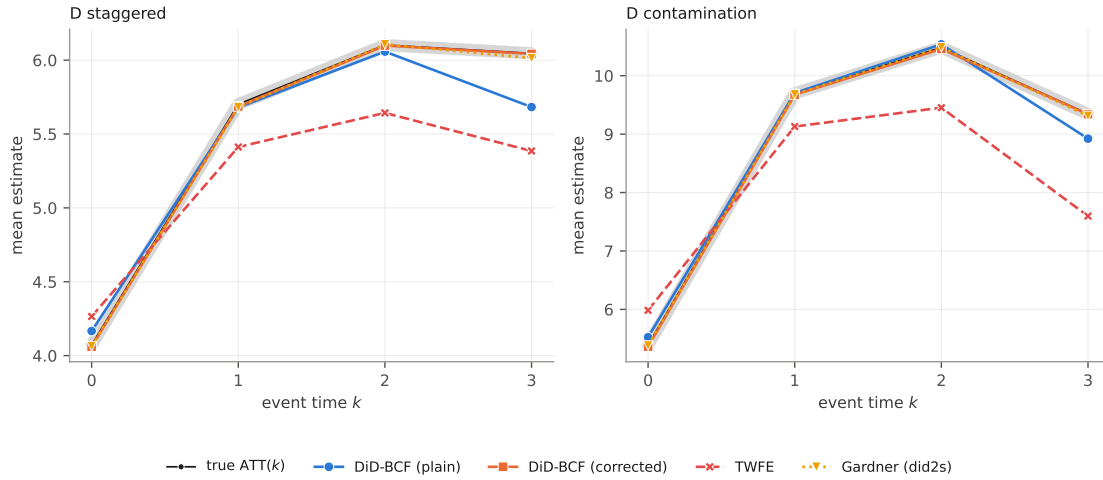


Figure 9: Event-study method comparison for the staggered designs.

Table 13: Post-correction B2 route comparison at $d = 1$, $N = 200$.

Method	Unit intercept	Summary	R	Bias	RMSE	Cov.95	Surface RMSE
DiD-BCF	no	plain	15	-0.006	0.095	1.000	0.248
DiD-BCF	no	corrected	15	-0.024	0.167	1.000	0.248
DiD-BCF	yes	plain	15	-0.009	0.091	1.000	0.241
DiD-BCF	yes	corrected	15	-0.031	0.159	1.000	0.241
Callaway-Sant'Anna	n/a	GATT(4, 4)	100	-0.031	0.221	0.980	1.069
Gardner	n/a	ES(0)	100	-0.047	0.179	0.980	1.072
DoubleML	n/a	GATT(4, 4)	100	-0.038	0.285	0.960	1.082
Synthetic DiD	n/a	ATT	100	-0.023	0.169	1.000	1.062
grf-DiD (Wang)	n/a	GATT(4, 4)	100	-0.038	0.222	0.850	0.809

Table 14: Post-correction B2 route sweep. Entries are bias, RMSE, 95% coverage, and CATT-surface RMSE.

Unit intercept	Summary	N	Bias	RMSE	Cov.95	Surface RMSE
no	plain	200	-0.006	0.095	1.000	0.248
no	corrected	200	-0.024	0.167	1.000	0.248
yes	plain	200	-0.009	0.091	1.000	0.241
yes	corrected	200	-0.031	0.159	1.000	0.241
no	plain	400	-0.013	0.140	1.000	0.183
no	corrected	400	-0.013	0.140	1.000	0.183
yes	plain	400	-0.009	0.140	1.000	0.175
yes	corrected	400	-0.009	0.140	1.000	0.175
no	plain	800	-0.012	0.095	1.000	0.168
no	corrected	800	-0.012	0.095	1.000	0.168
yes	plain	800	-0.012	0.095	1.000	0.159
yes	corrected	800	-0.012	0.095	1.000	0.159
no	plain	1600	-0.001	0.056	1.000	0.136
no	corrected	1600	-0.001	0.056	1.000	0.136
yes	plain	1600	0.003	0.056	1.000	0.125
yes	corrected	1600	0.003	0.056	1.000	0.125

Note. The values at $N = 400, 800, 1600$ are rounded summaries of the 15-replication route output. The corrected and plain point estimates are very close in this route file; the correction is more visible in posterior spread than in the displayed point metrics.

9.2 Interpretation and missing route cells

The unit-level random intercept is the preferred primary implementation for the canonical DGP once its full-grid evidence is available, because it targets the persistent unit-level component directly while preserving the two-way restriction. The no-unit-intercept structured fit remains necessary as a sensitivity analysis. The current route file does not support claims about unit-intercept performance in B1, B2 serial correlation, D staggered adoption, or $d = 2, 3$. Those cells must be run before choosing a final primary specification on the basis of route-specific surface or coverage results.

10 Overall interpretation and remaining work

The complete current structured grid supports four conclusions. First, the two-way-restricted DiD-BCF point estimates are stable across the canonical B1 and B2 designs, including the transformed nuisance axis and the nonlinear effect axis. Second, serial correlation at $d = 1$ is the setting in which the posterior correction has the clearest scalar inferential benefit, but that benefit does not survive the degree axis: at $d = 2$ and $d = 3$ the corrected event-study RMSE in the staggered designs exceeds pooled TWFE at every event time, so the correction cannot currently be recommended outside $d = 1$. Third, the structured fit handles the cohort-time heterogeneity in the D designs without the pooled TWFE contamination visible in the Goodman–Bacon decomposition, although the cell-level table shows that the plain fit pays for this with a concentrated bias and undercoverage in the late-adopting cohorts that the cell average hides. Fourth, both DiD-BCF specifications are correctly sized under the sharp null, the plain fit conservatively so at 0.01 to 0.05 and the corrected fit at 0.03 to 0.09, which is within Monte-Carlo error of nominal at 100 replications. On this axis the paper’s estimator is better behaved than grf-DiD, whose null rejection rate reaches 0.25.

The route audit adds a fifth, narrower conclusion. At $d = 1$ in the partial B2 ordinary sweep, the unit random intercept modestly improves the individual CATT surface and does not materially change scalar bias. This is exactly the comparison that should guide the next run, but it is not a substitute for running the unit-intercept route over every B1, B2, and D design and over $d = 1, 2, 3$.

The next computational block should therefore extend the route file in the following order. The B1 baseline, strong-confounder, serial-correlation, selection-on-observables, and null designs should be run first at all three degrees, because they identify whether the unit effect and correction interact with the degree axis. Finally, both D designs should be run with the unit intercept at all three degrees, with scalar GATT, event-study, and individual surface metrics retained. The sharp-null rejection rate, pointwise CATT coverage, and surface RMSE should be treated as explicit acceptance criteria, not as secondary diagnostics.

A Files used and reproducibility

The DGP and estimator definitions are in `Simulation_Studies_Revision/did_bcf_revision/dgps.py`, `Simulation_Studies_Revision/did_bcf_revision/structured.py`, and `didbcf_structured/design.py`. The metric definitions are in `Simulation_Studies_Revision/did_bcf_revision/metrics.py` and the posterior correction is in `Simulation_Studies_Revision/did_bcf_revision/posterior_correction.py`.

The complete-grid aggregates and generated tables were rebuilt with

```
Simulation_Studies_Revision/Results/analysis/aggregate_all.py
Simulation_Studies_Revision/Results/analysis/make_analysis.py
```

The generated figures used in this report are

```
figures/fig_sweep.pdf
figures/fig_sqrtn.pdf
figures/fig_calibration.pdf
figures/fig_surface.pdf
figures/fig_surface_sweep.pdf
figures/fig_event_study.pdf
figures/fig_ramp.pdf
figures/fig_es_methods.pdf
figures/fig_es_bias.pdf
```

The full structured input files are the CSV summaries under `DiD_BCF/`. The B2 ordinary and serial sweeps each contain 100 replications for every panel size $N \in \{50, 100, 200, 400, 800\}$ and degree $d \in \{1, 2, 3\}$, with both plain and corrected DiD-BCF summaries. The corresponding DoubleML files were unpacked from the eight B2 result archives under `DoubleML_Colab/` before the aggregates and tables were rebuilt. The route audit uses only `identification/routes_raw_B2_sweep_lin_1_full.csv`; its 15 replications per cell are kept separate from the full-grid aggregates. The report should be recompiled from this directory with two runs of `pdflatex`.